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SEMICONDUCTOR DEVICE AND METHOD FOR FABRICATING THE SAME

Abstract

Disclosed are highly integrated memory cells and a semiconductor device including the highly integrated memory cells. The semiconductor device includes a first pillar and a second pillar spaced apart from each other by a shield trench, each pillar including an inner side that defines inner walls of the shield trench and an outer side that faces the inner side; a shield gate formed in the shield trench; a first main gate formed on the outer side of the first pillar; a second main gate formed on the outer side of the second pillar; a bit line formed on lower portions of the first and second pillars; and a capacitor formed on an upper portion of each of the first and second pillars.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation-in-part application of U.S. patent application Ser. No. 17/844,941, filed on Jun. 21, 2022. The present application claims priority of Korean Patent Application No. 10-2021-0160342, filed on Nov. 19, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

1. Field

[0002] Embodiments of the present disclosure relate to a semiconductor device, and more particularly, to a semiconductor device including a bit line and a method for fabricating the semiconductor device.

2. Description of the Related Art

[0003] To fabricate highly integrated semiconductor devices, a transistor having a vertical channel has been proposed.

[0004] A vertical channel transistor using monocrystalline silicon as a channel has limitations in the degree of integration. Also, its electrical characteristics deteriorate due to gate-induced drain leakage (GIDL) occurring in an overlapping region between a source/drain and a gate and junction leakage of a PN junction.

SUMMARY

[0005] Embodiments of the present disclosure are directed to a highly integrated semiconductor device, and a method for fabricating the semiconductor device.

[0006] In accordance with an embodiment of the present disclosure, a semiconductor device includes a bit line; an oxide semiconductor pillar extending vertically from the bit line; a capacitor disposed over the oxide semiconductor pillar; and a word line disposed over a sidewall of the oxide semiconductor pillar, wherein the oxide semiconductor pillar includes a lower oxide semiconductor interface layer coupled to the bit line; an upper oxide semiconductor interface layer coupled to the capacitor; and an oxide semiconductor channel layer disposed between the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer.

[0007] In accordance with another embodiment of the present disclosure, a semiconductor device includes a substrate; a peripheral circuit portion disposed over the substrate; and a memory cell array including a bit line, a transistor, and a memory element that are vertically stacked over the peripheral circuit portion, wherein the transistor includes an oxide semiconductor channel layer disposed between the bit line and the memory element; a tapered vertical word line disposed over a sidewall of the oxide semiconductor channel layer; a lower oxide semiconductor interface layer between the bit line and the oxide semiconductor channel layer; and an upper oxide semiconductor interface layer disposed between the capacitor and the oxide semiconductor channel layer.

[0008] In accordance with yet another embodiment of the present disclosure, a method for fabricating a semiconductor device includes forming a bit line over a substrate; forming an oxide semiconductor pillar by stacking a lower interface layer, an oxide semiconductor channel layer, and an upper interface layer over the bit line in the recited order; forming a word line on a sidewall of the oxide semiconductor pillar; and forming a capacitor over the oxide semiconductor pillar.

[0009] In accordance with still another embodiment of the present disclosure, a semiconductor

device includes a first conductive line; an oxide semiconductor pillar extending vertically from the first conductive line; a memory element disposed over the oxide semiconductor pillar; and a second conductive line disposed over a sidewall of the oxide semiconductor pillar, wherein the oxide semiconductor pillar includes a lower oxide semiconductor interface layer coupled to the first conductive line; an upper oxide semiconductor interface layer coupled to the memory element; and an oxide semiconductor channel layer disposed between the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer.

[0010] In accordance with another embodiment of the present disclosure, a vertical channel transistor includes a lower oxide semiconductor interface layer; an upper oxide semiconductor interface layer; an oxide semiconductor channel layer extending vertically between the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer; and a tapered vertical gate disposed over a sidewall of the oxide semiconductor channel layer. The oxide semiconductor channel layer may include IGZO, ITZO or ZTO, and the lower oxide semiconductor interface layer and the upper oxide semiconductor interface layer may include indium-rich IGZO.

[0011] In accordance with another embodiment of the present disclosure, a semiconductor device includes a first pillar and a second pillar spaced apart from each other through a shield trench, each pillar including an inner side that defines inner walls of the shield trench and an outer side that faces the inner side; a shield gate formed in the shield trench; a first main gate formed on the outer side of the first pillar; a second main gate formed on the outer side of the second pillar; a bit line formed on lower portions of the first and second pillars; and a capacitor formed on an upper portion of each of the first and second pillars.

[0012] In accordance with another embodiment of the present disclosure, a semiconductor device includes a trench; a first row array of first oxide semiconductor pillars and a second row array of second oxide semiconductor pillars spaced apart from each other by the trench, each pillar including an inner side that defines inner sidewalls of the trench and an outer side that faces the inner side; a common gate extending along the first row array and the second row array, and formed in the trench while being formed on the inner sides of the first and second oxide semiconductor pillars; a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and a second gate extending along the outer sides of the second oxide semiconductor pillars of the second row array.

[0013] In accordance with another embodiment of the present disclosure, a method for fabricating a semiconductor device includes forming a bit line; forming a first row array of first oxide semiconductor pillars and a second row array of second oxide semiconductor pillars spaced apart from each other by a trench, each pillar including an inner side that defines inner sidewalls of the trench and an outer side that faces the inner side over the bit line; forming a common gate extending along the first row array and the second row array on the inner sides of the first and second oxide semiconductor pillars; forming a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and forming a second gate extending along the outer sides of the second oxide semiconductor pillars of the second row array.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1A is a schematic perspective view illustrating a semiconductor device.

[0015] FIG. 1B is a schematic plan view illustrating the semiconductor device of FIG. 1A.

[0016] FIG. 1C is a cross-sectional view taken along a line A-A' shown in FIG. 1B.

[0017] FIG. 1D is a cross-sectional view taken along a line B-B' shown in FIG. 1B.

[0018] FIG. 1E is a detailed cross-sectional view illustrating a capacitor.

[0019] FIGS. 2A to 2K are cross-sectional views illustrating a method for fabricating a

semiconductor device in accordance with an embodiment of the present disclosure.

[0020] FIGS. **3A** to **3E** are cross-sectional views illustrating a method for forming a capacitor.

[0021] FIG. **4A** is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure.

[0022] FIG. **4B** is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure.

[0023] FIG. **5** is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure.

[0024] FIGS. **6A** to **6C** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0025] FIGS. **7A** to **7G** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0026] FIGS. **8** to **10** are cross-sectional views illustrating semiconductor devices in accordance with other embodiments of the present disclosure.

[0027] FIGS. **11A** to **11C** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0028] FIGS. **12A** and **12B** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0029] FIGS. **13A** to **13D** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0030] FIGS. **14A** and **14B** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure.

[0031] FIG. **15A** is a schematic plan view illustrating a semiconductor device in accordance with an embodiment of the present disclosure.

[0032] FIG. **15B** is a cross-sectional view taken along a line A-A' shown in FIG. **15A**.

[0033] FIG. **15C** is a cross-sectional view taken along a line B-B' shown in FIG. **15A**.

[0034] FIG. **15D** is an enlarged view illustrating a portion **501** of FIG. **15A**.

[0035] FIGS. **16A** to **16I** illustrate a method for fabricating a semiconductor device in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0036] Embodiments of the present disclosure will be described below in more detail with reference to the accompanying drawings. The embodiments may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the present disclosure to those skilled in the art. Throughout this disclosure, like reference numerals refer to like parts throughout the various figures and embodiments of the present disclosure.

[0037] The drawings are not necessarily to scale and in some instances, proportions may have been exaggerated to clearly illustrate features of the embodiments. When a first layer is referred to as being “on” a second layer or “on” a substrate, it not only refers to a case where the first layer is formed directly on the second layer or the substrate but also a case where a third layer exists between the first layer and the second layer or the substrate.

[0038] FIG. **1A** is a schematic perspective view illustrating a semiconductor device. FIG. **1B** is a schematic plan view illustrating the semiconductor device. FIG. **1C** is a cross-sectional view taken along a line A-A' shown in FIG. **1B**. FIG. **1D** is a cross-sectional view taken along a line B-B' shown in FIG. **1B**. FIG. **1E** is a detailed cross-sectional view illustrating a capacitor **130**.

[0039] Referring to FIGS. **1A** to **1E**, the semiconductor device **100** may include a plurality of first conductive lines **110**, a plurality of oxide semiconductor pillars **120** vertically extending from the first conductive lines **110**, a plurality of memory elements **130** formed over the oxide semiconductor pillars **120**, and a plurality of second conductive lines **124** disposed over the

respective sidewalls of the oxide semiconductor pillars **120**. Each oxide semiconductor pillar **120** may include a lower interface layer **122** coupled to the first conductive line **110**, an upper interface layer **123** coupled to the memory element **130**, and an oxide semiconductor channel layer **121** disposed between the lower interface layer **122** and the upper interface layer **123**. The lower interface layer **122** and the upper interface layer **123** may include an oxide semiconductor layer. The lower interface layer **122** and the upper interface layer **123** may be referred to as a lower oxide semiconductor interface layer and an upper oxide semiconductor interface layer, respectively.

[0040] The semiconductor device **100** may include a Dynamic Random Access Memory (DRAM), and the first conductive lines **110** and the second conductive lines **124** may correspond to bit lines and word lines, respectively. The memory elements **130** may correspond to capacitors. Hereinafter, the first conductive lines **110** and the second conductive lines **124** may be simply referred to as bit lines **110** and word lines **124**, respectively, and the memory elements **130** may be simply referred to as capacitors **130**.

[0041] The semiconductor device **100** may include an oxide semiconductor channel layer **121**, a bit line **110** disposed at a lower level than the oxide semiconductor channel layer **121**, a capacitor **130** disposed at a higher level than the oxide semiconductor channel layer **121**, a tapered vertical word line **124** disposed over a sidewall of the oxide semiconductor channel layer **121**, a lower interface layer **122** between the bit line **110** and the oxide semiconductor channel layer **121**, and an upper interface layer **123** between the capacitor **130** and the oxide semiconductor channel layer **121**.

[0042] The semiconductor device **100** may be described in detail as follows.

[0043] The semiconductor device **100** may include a substrate **101**, a buffer layer **102** over the substrate **101**, a bit line **110** over the buffer layer **102**, a vertical channel transistor TR over the bit line **110**, and a capacitor **130** over the vertical channel transistor TR. The vertical channel transistor TR may include an oxide semiconductor pillar **120**, a tapered vertical word line **124** disposed over both sidewalls of the oxide semiconductor pillar **120**, and a gate dielectric layer **125** between the oxide semiconductor pillar **120** and the tapered vertical word line **124**. The oxide semiconductor pillar **120** may include the oxide semiconductor channel layer **121**, the lower interface layer **122** between the oxide semiconductor channel layer **121** and the bit line **110**, and the upper interface layer **123** between the oxide semiconductor channel layer **121** and the capacitor **130**. A barrier layer **111** may be disposed between the bit line **110** and the lower interface layer **122**.

[0044] The semiconductor device **100** may further include first and second contact plugs **126** and **127** between the upper interface layer **123** and the capacitor **130**.

[0045] The substrate **101** may be a material suitable for semiconductor processing. The substrate **101** may include a semiconductor substrate. The substrate **101** may be formed of a silicon-containing material. The substrate **101** may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate **101** may include other semiconductor materials such as germanium. The substrate **101** may include a III/V group semiconductor substrate, for example, a compound semiconductor substrate, such as GaAs. The substrate **101** may include a Silicon-On-Insulator (SOI) substrate.

[0046] The buffer layer **102** may include silicon oxide, silicon nitride, or a combination thereof. To reduce parasitic capacitance, the buffer layer **102** may be formed of silicon oxide. For example, the buffer layer **102** may include tetra ethyl ortho silicate (TEOS).

[0047] The bit line **110** may extend in a first direction D1 over the buffer layer **102**. The bit line **110** may include a metal-based material. The bit line **110** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The bit line **110** may have a thickness of approximately 100 to 400 Å. The bit line **110** may include a tungsten layer.

[0048] The barrier layer **111** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier layer **111** may include titanium nitride, molybdenum, or ruthenium. The barrier layer **111** may have a thickness of approximately 10 to 50 Å. For example, the barrier layer **111**

may include a titanium nitride layer.

[0049] The lower interface layer **122** and the upper interface layer **123** may include an oxide semiconductor material having a lower resistance than the oxide semiconductor channel layer **121**. The lower interface layer **122** and the upper interface layer **123** may include a metallic-rich oxide semiconductor material, and the oxide semiconductor channel layer **121** may include an oxygen-rich oxide semiconductor material. For example, the oxide semiconductor channel layer **121** may include indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO), or zinc tin oxide (ZTO), and the lower interface layer **122** and the upper interface layer **123** may include indium-rich IGZO. Indium-rich IGZO may refer to a material with a higher indium content than gallium (Ga) and zinc (Zinc) in IGZO, for example, the content of indium may be approximately 40% or more.

[0050] The oxide semiconductor channel layer **121** may include an oxide semiconductor material. The oxide semiconductor channel layer **121** may contain indium. The oxide semiconductor channel layer **121** may include IGZO. The oxide semiconductor channel layer **121** may be formed to have a thickness of approximately 200 to 1000 Å.

[0051] The lower interface layer **122** may include an oxide semiconductor material. The lower interface layer **122** may contain indium. The lower interface layer **122** may include an indium-rich oxide semiconductor material. For example, the lower interface layer **122** may include indium-rich IGZO. The lower interface layer **122** may be formed to have a thickness of approximately 10 to 50 Å.

[0052] The upper interface layer **123** may include an oxide semiconductor material. The upper interface layer **123** may contain indium. The upper interface layer **123** may include an indium-rich oxide semiconductor material. For example, the upper interface layer **123** may include indium-rich IGZO. The upper interface layer **123** may be formed to have a thickness of approximately 10 to 50 Å.

[0053] As described above, the oxide semiconductor pillar **120** may extend vertically in a third direction **D3** over the bit line **110**. The oxide semiconductor pillar **120** may be vertically stacked over the bit line **110** in an order of the lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123**. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include an oxide semiconductor material. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include IGZO, but the lower interface layer **122** and the upper interface layer **123** may have a higher indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may be IGZO, and the lower interface layer **122** and the upper interface layer **123** may be indium-rich IGZO. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may be referred to as an active pillar.

[0054] A tapered vertical word line **124** having a double structure may be disposed over a sidewall of the oxide semiconductor channel layer **121**. The tapered vertical word line **124** and the bit line **110** may extend in directions crossing each other. The tapered vertical word line **124** may have a reverse tapered shape. The reverse tapered shape may refer to a shape the width of the bottom portion of which becomes gradually narrower than the width of the upper portion. For example, the tapered vertical word line **124** may include a lower level portion **124L** which is adjacent to the bit line **110** and an upper level portion **124U** which is adjacent to the capacitor **130**. The thickness of the lower level portion **124L** in the first direction **D1** may be thinner than the thickness of the upper level portion **124U**. The lower level portion **124L** may be adjacent to the lower interface layer **122**, and the upper level portion **124U** may be adjacent to the upper interface layer **123**. The lower level portion **124L** and the upper level portion **124U** may be formed of the same material. The lower level portion **124L** and the bit line structure **BL** may be spaced apart from each other with a space. The lower level portion **124L** of the tapered vertical word line **124** and the bit line structure **BL** may not contact each other.

[0055] The tapered vertical word line **124** may include a metal-based material. The tapered vertical word line **124** may include a metal, a metal nitride, or a combination thereof. The tapered vertical word line **124** may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0056] A gate dielectric layer **125** may be formed between the tapered vertical word line **124** and the oxide semiconductor pillar **120**. The gate dielectric layer **125** may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The gate dielectric layer **125** may be disposed between the oxide semiconductor pillar **120** and the tapered vertical word line **124**. The gate dielectric layer **125** may include a horizontal portion that extends to be disposed between the lower level portion **124L** of the tapered vertical word line **124** and the bit line **110**, and the horizontal portion of the gate dielectric layer **125** may directly contact the barrier layer **111**. The oxide semiconductor pillar **120**, the gate dielectric layer **125**, and the tapered vertical word line **124** may form the vertical channel transistor TR. The tapered vertical word line **124** may be referred to as a tapered vertical gate.

[0057] The contact plugs **126** and **127** may include a first contact plug **126** and a second contact plug **127** over the first contact plug **126**. The first contact plug **126** may directly contact the upper interface layer **123**, and the second contact plug **127** may directly contact the capacitor **130**. The first contact plug **126** and the second contact plug **127** may vertically overlap with each other. The first contact plug **126** and the second contact plug **127** may include a metal-based material. The first contact plug **126** and the second contact plug **127** may include a metal, a metal nitride, or a combination thereof. The first contact plug **126** and the second contact plug **127** may be formed of the same metal-based material. According to another embodiment of the present disclosure, the first contact plug **126** and the second contact plug **127** may be formed of different metal-based materials. Sidewalls of the first contact plugs **126** may be surrounded by the gate dielectric layer **125**.

[0058] The capacitor **130** may include a lower electrode **132**, a dielectric layer **135**, and an upper electrode **136**. The lower electrode **132** may be formed over the second contact plug **127**. The lower electrode **132** may have a pillar shape. The lower electrodes **132** may be supported by the supporters **133** and **134**. A sidewall of the bottom portion of the lower electrode **132** may contact the etch stop layer **131**. According to another embodiment of the present disclosure, the lower electrodes **132** may have a cylindrical shape.

[0059] Although not illustrated, dielectric pillars may be disposed between the oxide semiconductor pillars **120** in the direction of the A-A'.

[0060] FIGS. 2A to 2K are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with an embodiment of the present disclosure.

[0061] Referring to FIG. 2A, a buffer layer **12** may be formed over a substrate **11**. The substrate **11** may be a material appropriate for semiconductor processing. The substrate **11** may include a semiconductor substrate. The substrate **11** may be formed of a silicon-containing material. The substrate **11** may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate **11** may include other semiconductor materials, such as germanium. The substrate **11** may include a III/V-group semiconductor substrate, for example, a compound semiconductor substrate, such as GaAs. The substrate **11** may include a Silicon-On-Insulator (SOI) substrate. The buffer layer **12** may include a dielectric material. The buffer layer **12** may include silicon oxide, silicon nitride, or a combination thereof.

[0062] A conductive layer **13A** may be formed over the buffer layer **12**. The conductive layer **13A** may include a metal-based material. The conductive layer **13A** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The conductive layer **13A** may be formed by atomic layer deposition (ALD), physical vapor deposition (PVD), or chemical vapor deposition (CVD).

The conductive layer **13A** may be formed to have a thickness of approximately 100 to 400 Å. For example, as for the conductive layer **13A**, a tungsten layer may be deposited to have a thickness of approximately 200 Å by physical vapor deposition (PVD). To reduce parasitic capacitance between the substrate **11** and the conductive layer **13A**, the buffer layer **12** may be formed of silicon oxide. For example, the buffer layer **12** may include tetra ethyl ortho silicate (TEOS).

[0063] A barrier material layer **14A** may be formed over the conductive layer **13A**. The barrier material layer **14A** may include a metal-based material. The barrier material layer **14A** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier material layer **14A** may include titanium nitride, molybdenum, or ruthenium. The barrier material layer **14A** may be formed by atomic layer deposition (ALD), physical vapor deposition (PVD), or chemical vapor deposition (CVD). The barrier material layer **14A** may be formed to have a thickness of approximately 10 to 50 Å. For example, as the barrier material layer **14A**, titanium nitride may be deposited to have a thickness of approximately 20 Å by physical vapor deposition (PVD).

[0064] An oxide semiconductor stack may be formed over the barrier material layer **14A**. For example, the oxide semiconductor stack may include a lower interface material layer **15A**, a channel material layer **16A**, and an upper interface material layer **17A**.

[0065] A lower interface material layer **15A** may be formed over the barrier material layer **14A**. The lower interface material layer **15A** may include a conductive material. The lower interface material layer **15A** may include an oxide semiconductor material. The lower interface material layer **15A** may contain indium. The lower interface material layer **15A** may include an indium-rich oxide semiconductor material. For example, the lower interface material layer **15A** may include indium-rich IGZO. The lower interface material layer **15A** may be formed to have a thickness of approximately 10-50 Å.

[0066] A channel material layer **16A** may be formed over the lower interface material layer **15A**. The channel material layer **16A** may include a conductive material. The channel material layer **16A** may include an oxide semiconductor material. The channel material layer **16A** may contain indium. The channel material layer **16A** may include IGZO. The channel material layer **16A** may be formed to have a thickness of approximately 200 to 1000 Å.

[0067] An upper interface material layer **17A** may be formed over the channel material layer **16A**. The upper interface material layer **17A** may include a conductive material. The upper interface material layer **17A** may include an oxide semiconductor material. The upper interface material layer **17A** may contain indium. The upper interface material layer **17A** may include an indium-rich oxide semiconductor material. For example, the upper interface material layer **17A** may include indium-rich IGZO. The upper interface material layer **17A** may be formed to have a thickness of approximately 10 to 50 Å.

[0068] As described above, the lower interface material layer **15A**, the channel material layer **16A**, and the upper interface material layer **17A** may be vertically stacked over the barrier material layer **14A**. The lower interface material layer **15A**, the channel material layer **16A**, and the upper interface material layer **17A** may all include an oxide semiconductor material. The lower interface material layer **15A**, the channel material layer **16A**, and the upper interface material layer **17A** may all include IGZO, but the lower interface material layer **15A** and the upper interface material layer **17A** may have a higher indium concentration than the channel material layer **16A**. The channel material layer **16A** may be IGZO, and the lower interface material layer **15A** and the upper interface material layer **17A** may be indium-rich IGZO. Since the lower interface material layer **15A** and the upper interface material layer **17A** contain a high concentration of indium, its resistance may be reduced lower than that of the channel material layer **16A**. Also, channel seamless interconnection of the channel material layer **16A** may be possible.

[0069] Subsequently, a sacrificial layer **18A** may be formed over the upper interface material layer **17A**. The sacrificial layer **18A** may include a stack of different materials. The sacrificial layer **18A** may include silicon nitride.

[0070] Referring to FIG. 2B, sacrificial lines **18** may be formed over the upper interface material layer **17A**. The sacrificial lines **18** may be formed by etching the sacrificial layer **18A**. The sacrificial lines **18** may serve to protect the lower interface material layer **15A**, the channel material layer **16A**, and the upper interface material layer **17A** from the subsequent processes. For example, the sacrificial lines **18** may be used as an etch barrier during an etching process of the lower interface material layer **15A**, the channel material layer **16A**, and the upper interface material layer **17A**.

[0071] Each of the sacrificial lines **18** may include a stack of different materials. The sacrificial lines **18** may include silicon nitride. The etching process of the sacrificial layer **18A** to form the sacrificial lines **18** may include a double patterning process.

[0072] Subsequently, the upper interface material layer **17A**, the channel material layer **16A**, and the lower interface material layer **15A** may be etched by using the sacrificial lines **18** as an etch barrier, and then the barrier material layer **14A** and the conductive layer **13A** may be etched.

[0073] A plurality of line structures **19L** and first trenches **19T** may be formed over the buffer layer **12** by a series of the etching processes described above. Each of the line structures **19L** may include a stack of a bit line **13**, a bit line barrier layer **14**, a lower interface layer **15B**, a channel material **16B**, an upper interface layer **17B**, and a sacrificial line **18**. The bit line barrier layer **14** may be formed by etching the barrier material layer **14A**, and the bit line **13** may be formed by etching the conductive layer **13A**. The lower interface layer **15B**, the channel material **16B**, and the upper interface layer **17B** may be formed by etching the lower interface material layer **15A**, the channel material layer **16A**, and the upper interface material layer **17A**, respectively. The first trenches **19T** may be disposed between the line structures **19L**. The stack of the lower interface layer **15B**, the channel material **16B**, and the upper interface layer **17B** may be referred to as an 'oxide semiconductor line'.

[0074] Referring to FIG. 2C, dielectric lines **20** may be formed between the line structures **19L**. The dielectric lines **20** may include a dielectric material. For example, the dielectric lines **20** may include silicon oxide, silicon nitride, silicon carbon oxide (SiCO), spin-on-dielectric, or a combination thereof. After a dielectric material is deposited to fill the first trenches **19T** between the line structures **19L** to form the dielectric lines **20**, a planarization process of the dielectric material may be performed. The dielectric lines **20** may fill the first trenches **19T**, respectively. To form the dielectric lines **20**, a planarization process may be performed after a silicon nitride, silicon carbon oxide, and a spin-on dielectric layer are sequentially formed.

[0075] Referring to FIG. 2D, a portion of the line structures **19L** may be selectively etched to form the sacrificial pillars **18P** and the oxide semiconductor pillars **21P**. A bit line barrier layer **14** and a bit line **13** may be disposed below the oxide semiconductor pillars **21P**.

[0076] Each of the oxide semiconductor pillars **21P** may include a stack of a lower interface layer **15**, a channel layer **16**, and an upper interface layer **17**. The lower interface layer **15**, the channel layer **16**, and the upper interface layer **17** may be formed by etching the lower interface layer **15B**, the channel material **16B**, and the upper interface layer **17B**, respectively. A sacrificial pillar **18P** may be formed over the upper interface layer **17**.

[0077] Second trenches **22** may be formed between the oxide semiconductor pillars **21P**. The bottom surfaces of the second trenches **22** may expose the top surface of the bit line barrier layer **14**. The first trenches **19T** and the second trenches **22** may intersect with each other. The first trenches **19T** may be deeper than the second trenches **22**. In the direction of the line B-B', the dielectric lines **20** may be cut by the second trenches **22**. Hereinafter, the dielectric lines **20** may be simply referred to as 'dielectric pillars **20**'. The dielectric pillars **20** may be disposed between the oxide semiconductor pillars **21P** in the direction of the line A-A'.

[0078] Each of the oxide semiconductor pillars **21P** may include first to fourth sidewalls SW1 to SW4. A first sidewall SW1 and a second sidewall SW2 of the individual oxide semiconductor pillar **21P** may be exposed by the second trenches **22**, and a third sidewall SW3 and a fourth sidewall

SW4 of the individual oxide semiconductor pillar 21P may not be exposed by the dielectric pillars 20. The sacrificial pillar 18P may also include exposed sidewalls and non-exposed sidewalls just as the oxide semiconductor pillar 21P.

[0079] Referring to FIG. 2E, a gate dielectric layer 23 may be formed over the exposed first and second sidewalls SW1 and SW2 of the oxide semiconductor pillars 21P. The gate dielectric layer 23 may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The high-k material may include a material having a higher dielectric constant than that of silicon oxide. For example, the high-k material may include a material having a dielectric constant which is greater than approximately 3.9. In another embodiment, the high-k material may include a material having a dielectric constant which is greater than approximately 10. In another embodiment, the high-k material may include a material having a dielectric constant of approximately 10 to 30. The high-k material may include at least one metallic element. The high-k material may include a hafnium-containing material. The hafnium-containing material may include hafnium oxide, hafnium silicon oxide, hafnium silicon oxynitride, or a combination thereof. According to another embodiment of the present disclosure, the high-k material may include lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, zirconium silicon oxynitride, aluminum oxide, or a combination thereof. As for the high-k material, other known high-k materials may be selectively used. The gate dielectric layer 23 may include a metal oxide. The gate dielectric layer 23 may be formed conformally over the first and second sidewalls SW1 and SW2 of the oxide semiconductor pillar 21P. The gate dielectric layer 23 may conformally cover the exposed sidewalls and the top surface of the sacrificial pillar 18P. The gate dielectric layer 23 and the lower interface layer 15 may have the same thickness. According to another embodiment of the present disclosure, the gate dielectric layer 23 may be thinner than the lower interface layer 15.

[0080] A word line conductive layer 24A may be formed over the gate dielectric layer 23. The word line conductive layer 24A may be conformally formed over the gate dielectric layer 23. The word line conductive layer 24A may include a metal-based material. The word line conductive layer 24A may include a metal, a metal nitride, or a combination thereof. The word line conductive layer 24A may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0081] Spacers 25 may be formed over the word line conductive layer 24A. The spacers 25 may include an oxide. To form the spacers 25, an etch-back process may be performed after depositing silicon oxide over the word line conductive layer 24A. The upper surface of the spacers 25 may be disposed at a lower level than the upper surface of the sacrificial pillars 18P.

[0082] Referring to FIG. 2F, tapered vertical word lines 24 may be formed. To form the tapered vertical word lines 24, the word line conductive layer 24A may be selectively etched by using the spacers 25 as an etch barrier. The spacers 25 may serve to protect the tapered vertical word lines 24 during an etching process of the word line conductive layer 24A. In the etching process of the word line conductive layer 24A, an etch-back process and a wet etching process may be sequentially performed. Non-tapered vertical word lines may be formed by an etch-back process, and the tapered vertical word lines 24 may be formed by the subsequent wet etching process. The bottom portion of the tapered vertical word lines 24 may become thin by the wet etching process. The tapered vertical word lines 24 may correspond to the tapered vertical word lines 124 of FIGS. 1A, 1B, and 1D, and the bottom portion of the tapered vertical word lines 24 may correspond to the lower level portion 124L.

[0083] Referring to FIG. 2G, the spacers 25 may be removed. Each of the tapered vertical word lines 24 may have a double structure and the tapered vertical word lines 24 may be respectively formed on the first and second sidewalls of the oxide semiconductor pillars 21P. Referring to FIGS. 1A, 1B, and 1D, the tapered vertical word lines 24 may extend in the second direction D2, and the bit lines 13 may extend in the first direction D1. The oxide semiconductor pillars 21P and the

dielectric pillars **20** may be alternately disposed in the second direction **D2**, and the tapered vertical word lines **24** may extend along the exposed sidewalls of the oxide semiconductor pillars **21P** and the dielectric pillar **20**.

[0084] Referring to FIG. **2H**, an inter-layer dielectric layer **26** may be formed over the tapered vertical word line **24**. The inter-layer dielectric layer **26** may be planarized to expose the upper surfaces of the sacrificial pillars **18P**. The inter-layer dielectric layer **26** may include silicon oxide, such as a spin-on dielectric layer (SOD). The sacrificial pillars **18P** may serve as an etch stop layer during a planarization process of the inter-layer dielectric layer **26**.

[0085] Subsequently, the sacrificial pillars **18P** may be selectively removed. Accordingly, hole-shaped recesses **18R** may be formed. The hole-shaped recesses **18R** may selectively expose the surfaces of the upper interface layers **17**. The sacrificial pillars **18P** may be removed by using a wet etching process.

[0086] Referring to FIG. **2I**, first contact plugs **27** filling the hole-shaped recesses **18R** may be formed. The first contact plugs **27** may directly contact the upper interface layers **17**. The first contact plugs **27** may include a metal, a metal nitride, or a combination thereof. For example, to form the first contact plugs **27**, titanium nitride may be deposited to fill the hole-shaped recesses **18R** and then titanium nitride may be planarized to expose the surface of the inter-layer dielectric layer **26**.

[0087] Referring to FIG. **2J**, second contact plugs **28** may be formed over the first contact plugs **27**. The first contact plugs **27** and the second contact plugs **28** may partially overlap with each other. The second contact plugs **28** may include a metal-based material. The second contact plugs **28** may include a metal, a metal nitride, or a combination thereof. For example, the second contact plugs **28** may include tungsten. According to another embodiment of the present disclosure, the first contact plugs **27** and the second contact plugs **28** may be formed of the same metal-based material.

[0088] A metal-based material may be deposited and etched to form the second contact plugs **28**.

[0089] Referring to FIG. **2K**, a spacer material layer **29** may be formed over the second contact plugs **28**. The spacer material layer **29** may include silicon nitride. The spacer material layer **29** may be formed between the neighboring second contact plugs **28**.

[0090] A capacitor **30** may be formed over the second contact plug **28**.

[0091] FIGS. **3A** to **3E** are cross-sectional views illustrating a method for forming the capacitor **30**. Hereinafter, the structures formed before the formation of the second contact plugs **28** are omitted.

[0092] Referring to FIG. **3A**, an etch stop layer **31** may be formed over the second contact plugs **28** and the spacer material layer **29**. A first mold layer **32**, a first supporter layer **33**, a second mold layer **34**, and a second supporter layer **35** may be sequentially formed over the etch stop layer **31**.

[0093] The etch stop layer **31** may be formed of a material having an etch selectivity with respect to the first mold layer **32**. The etch stop layer **31** may include silicon nitride or silicon oxynitride. The first mold layer **32** may include a dielectric material. The first mold layer **32** may be formed of silicon oxide (SiO₂). The first mold layer **32** may be formed to be thicker than the first supporter layer **33**. The first mold layer **32** may be formed by using a deposition process, such as chemical vapor deposition (CVD), atomic layer deposition (ALD), or physical vapor deposition (PVD). The first mold layer **32** may include silicon oxide doped with phosphorus or silicon oxide doped with boron. The first mold layer **32** may include USG (Undoped Silicate Glass), PSG (Phosphorous Silicate Glass), BSG (Boron Silicate Glass), BPSG (Boron Phosphorous Silicate Glass), FSG (Fluorine Silicate Glass), or a combination thereof. Phosphorus-doped silicon oxide and boron-doped silicon oxide may be readily removed during the subsequent process because the etching rate with respect to an etching solution is high.

[0094] The first supporter layer **33** may be formed of a material having an etch selectivity with respect to the first mold layer **32** and the second mold layer **34**. The first supporter layer **33** may include silicon nitride or silicon carbon nitride (SiCN).

[0095] The second mold layer **34** may include a dielectric material. The second mold layer **34** may

be formed of silicon oxide (SiO₂). The second mold layer **34** may be formed to be thicker than the first supporter layer **33**. The second mold layer **34** may be formed by a deposition process, such as chemical vapor deposition (CVD), atomic layer deposition (ALD), or physical vapor deposition (PVD). The second mold layer **34** may include phosphorus-doped silicon oxide or boron-doped silicon oxide. The second mold layer **34** may include USG, PSG, BSG, BPSG, FSG, or a combination thereof. The first mold layer **32** and the second mold layer **34** may be formed of the same material or different materials.

[0096] According to another embodiment of the present disclosure, the first mold layer **32** and the second mold layer **34** may be formed of a silicon material, such as amorphous silicon or polysilicon.

[0097] The second supporter layer **35** may be formed of a material having an etch selectivity with respect to the second mold layer **34**. The second supporter layer **35** may include silicon nitride or silicon carbon nitride (SiCN).

[0098] The first supporter layer **33** and the second supporter layer **35** may be formed of the same material or different materials. Both of the first supporter layer **33** and the second supporter layer **35** may be formed of silicon nitride. According to another embodiment of the present disclosure, the first supporter layer **33** may be formed of silicon nitride, and the second supporter layer **35** may be formed of silicon carbon nitride. The second supporter layer **35** may be thicker than the first supporter layer **33**.

[0099] According to another embodiment of the present disclosure, another supporter layer may be further formed. For example, the supporter structure may be a multi-level supporter layer structure.

[0100] Subsequently, an opening **36** may be formed. To form the opening **36**, the second supporter layer **35**, the second mold layer **34**, the first supporter layer **33**, and the first mold layer **32** may be sequentially etched by using a mask layer (not shown) as an etch barrier. An etching process for forming the opening **36** may stop at the etch stop layer **31**. To form the opening **36**, a dry etching process, a wet etching process, or a combination thereof may be used. The opening **36** may be referred to as a hole in which a lower electrode (or a storage node) is to be formed.

[0101] Subsequently, the etch stop layer **31** may be etched to expose the upper surface of the second contact plug **38** below the opening **36**.

[0102] Referring to FIG. 3B, a lower electrode **37** may be formed in the opening **36**. The lower electrode **37** may fill the inside of the opening **36**. The lower electrode **37** may include polysilicon, a metal, a metal nitride, a conductive metal oxide, a metal silicide, a noble metal, or a combination thereof. The lower electrode **37** may include at least one of titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta₂N₃), titanium aluminum nitride (TiAlN), tungsten (W), tungsten nitride (WN), ruthenium (Ru), ruthenium oxide (RuO₂), iridium (Ir), iridium oxide (IrO₂), platinum (Pt), and a combination thereof. The lower electrode **37** may include titanium nitride (TiN). The lower electrode **37** may include titanium nitride (ALD-TiN) which is formed by atomic layer deposition (ALD). According to another embodiment of the present disclosure, the lower electrode **37** may include a hybrid structure of a titanium nitride cylinder and a polysilicon pillar.

[0103] Referring to FIG. 3C, the second supporter layer **35**, the second mold layer **34**, and the first supporter layer **33** may be sequentially etched. As a result, a supporter opening **38** exposing the first mold layer **32** may be formed, and an upper-level supporter **35S** and a lower-level supporter **33S** may be formed.

[0104] The upper level supporter **35S** and the lower level supporter **33S** may contact the outer wall of the lower electrode **37**. The upper-level supporter **35S** and the lower-level supporter **33S** may prevent the lower electrodes **37** from collapsing in the subsequent process of removing the second mold layer **34** and the first mold layer **32**.

[0105] Referring to FIG. 3D, the second mold layer **34** and the first mold layer **32** may be removed through the supporter opening **38**. The first and second mold layers **32** and **34** may be removed by

a wet dip-out process. The wet dip-out process for removing the first and second mold layers **32** and **34** may be performed by using an etching solution capable of selectively removing the first and second mold layers **32** and **34**. When the first and second mold layers **32** and **34** include silicon oxide, the first and second mold layers **32** and **34** may be removed by a wet etching process using hydrofluoric acid (HF).

[0106] Referring to FIG. **3E**, a dielectric layer **39** may be formed over the lower electrode **37** and the lower and upper level supporters **33S** and **35S**. The dielectric layer **39** may include a high-k material having a higher dielectric constant than silicon oxide. The high-k material may include hafnium oxide (HfO₂), zirconium oxide (ZrO₂), aluminum oxide (Al₂O₃), titanium oxide (TiO₂), tantalum oxide (Ta₂O₅), niobium oxide (Nb₂O₅) or strontium titanium oxide (SrTiO₃). According to another embodiment of the present disclosure, the dielectric layer **39** may be formed of a composite layer including two or more layers of the above-mentioned high-k materials. According to the embodiment of the present disclosure, the dielectric layer **39** may be formed of a zirconium oxide-based material having excellent leakage current characteristics while sufficiently lowering the equivalent oxide thickness (EOT). For example, it may include a ZAZ (ZrO₂/Al₂O₃/ZrO₂) stack. According to another embodiment of the present disclosure, the dielectric layer **27** may include a

TiO₂/ZrO₂/Al₂O₃/ZrO₂ stack, a TiO₂/HfO₂/Al₂O₃/HfO₂ stack, a Ta₂O₅/ZrO₂/Al₂O₃/ZrO₂ stack, or a Ta₂O₅/HfO₂/Al₂O₃/HfO₂ stack.

[0107] Subsequently, an upper electrode **40** may be formed over the dielectric layer **39**. The upper electrode **40** may fill the space between the neighboring lower electrodes **37**. The upper electrode **40** may extend to cover the upper portions of the lower electrodes **37**. The upper electrode **40** may include a conductive material. The upper electrode **40** may be stacked (reference numerals omitted) in the order of a liner electrode, a gap-fill electrode, and a low-resistance electrode. The liner electrode of the upper electrode **40** may include titanium nitride, and the gap-fill electrode of the upper electrode **40** may include silicon germanium. The low resistance electrode of the upper electrode **40** may include tungsten or tungsten nitride.

[0108] FIG. **4A** is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure. The semiconductor device **200** of FIG. **4A** may be similar to the semiconductor device **100** shown in FIGS. **1A** to **1E**. Hereinafter, for the detailed descriptions of the constituent elements which also appear in FIGS. **1A** to **1E**, the description of FIGS. **1A** to **1E** may be referred to.

[0109] Referring to FIG. **4A**, the semiconductor device **200** may include a peripheral circuit portion PERI including a substrate **201**, and a memory cell array MCA over the peripheral circuit portion PERI. The memory cell array MCA may include a bit line **110**, a transistor TR, and a capacitor **130**. The transistor TR may include a vertical channel transistor. The memory cell array MCA may include a plurality of memory cells sharing the bit line **110**.

[0110] The transistor TR may include an oxide semiconductor pillar **120** which is disposed between the bit line **110** and the capacitor **130**, a tapered vertical word line **124** which is disposed over a sidewall of the oxide semiconductor pillar **120**, and a gate dielectric layer **125** disposed between the oxide semiconductor pillar **120** and the tapered vertical word line **124**. The oxide semiconductor pillar **120** may include a lower interface layer **122**, an oxide semiconductor channel layer **121**, and an upper interface layer **123**. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include an oxide semiconductor material. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include IGZO, but the lower interface layer **122** and the upper interface layer **123** may have a greater indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may be IGZO, and the lower interface layer **122** and the upper

interface layer **123** may be indium-rich IGZO.

[0111] The tapered vertical word line **124** may include a lower level portion **124L** and an upper level portion **124U**.

[0112] The bit line **110** and a peripheral transistor PERI_TR of the peripheral circuit portion PERI may be coupled to each other through a metal interconnection MLM. The uppermost layer of the metal interconnection MLM may pass through a buffer layer **102** to be coupled to the bit line **110**.

[0113] The semiconductor device **200** may include a peripheral-under-cell (PUC) structure in which the memory cell array MCA is formed over the peripheral circuit portion PERI.

[0114] FIG. **4B** is a cross-sectional view illustrating a semiconductor device **200M** in accordance with another embodiment of the present disclosure. The semiconductor device **200M** of FIG. **4B** may be similar to the semiconductor device **100** shown in FIGS. **1A** to **1E**. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. **1A** to **1E**, descriptions of FIGS. **1A** to **1E** may be referred to.

[0115] Referring to FIG. **4B**, the semiconductor device **200M** may be an array of memory cells including a bit line **110**, an oxide semiconductor pillar **120**, and a capacitor **130**. For example, the lower level memory cell array MCA_L and the upper level memory cell array MCA_U may be vertically stacked. Each of the memory cells of the lower level memory cell array MCA_L and the upper level memory cell array MCA_U may include a bit line **110**, a transistor TR, and a capacitor **130**. The transistor TR may include a vertical channel transistor. The transistor TR may include an oxide semiconductor pillar **120** which is disposed between the bit line **110** and the capacitor **130**, a tapered vertical word line **124** which is disposed over a sidewall of the oxide semiconductor pillar **120**, and a gate dielectric layer **125** disposed between the oxide semiconductor pillar **120** and the tapered vertical word line **124**. The oxide semiconductor pillar **120** may include a lower interface layer **122**, an oxide semiconductor channel layer **121**, and an upper interface layer **123**. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include an oxide semiconductor material. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include IGZO, but the lower interface layer **122** and the upper interface layer **123** may have a higher indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may be IGZO, and the lower interface layer **122** and the upper interface layer **123** may be indium-rich IGZO. The tapered vertical word line **124** may include a lower level portion **124L** and an upper level portion **124U**.

[0116] The upper electrode **136** of the lower level memory cell array MCA_L may directly contact the buffer layer **102** of the upper level memory cell array MCA_U.

[0117] The lower level memory cell array MCA_L and the upper level memory cell array MCA_U may be stacked without wafer bonding.

[0118] FIG. **5** is a cross-sectional view illustrating a semiconductor device in accordance with another embodiment of the present disclosure. The semiconductor device **300** of FIG. **5** may be similar to the semiconductor device **100** of FIGS. **1A** to **1E**. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. **1A** to **1E**, descriptions of FIGS. **1A** to **1E** may be referred to.

[0119] Referring to FIG. **5**, the semiconductor device **300** may include a bit line **110**, a vertical channel transistor TR, and a capacitor **130** that are disposed vertically in the third direction D3. The vertical channel transistor TR may include an oxide semiconductor pillar **120** disposed between the bit line **110** and the capacitor **130**, a tapered vertical word line **124** disposed over a sidewall of the oxide semiconductor pillar **120**, and a gate dielectric layer **125** disposed between the oxide semiconductor pillar **120** and the tapered vertical word line **124**. The oxide semiconductor pillar **120** may include a lower interface layer **122**, an oxide semiconductor channel layer **121**, and an upper interface layer **123**. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include an oxide semiconductor material. The lower

interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include IGZO, but the lower interface layer **122** and the upper interface layer **123** may have a higher indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may be IGZO, and the lower interface layer **122** and the upper interface layer **123** may be indium-rich IGZO.

[0120] The semiconductor device **300** may further include a dummy plate **210** below the bit line **110**. The dummy plate **210** may include a metal-based material. The buffer layer **102** may be disposed between the dummy plate **210** and the bit line **110**. Although parasitic capacitance may be increased due to the dummy plate **210**, coupling noise may be improved because the coupling capacitance ratio between the bit lines **110** is decreased. Referring to FIGS. **1B** and **5**, the coupling capacitance ratio between the bit lines **110** that are spaced apart from each other in the second direction **D2** may decrease by the dummy plate **210**. The optimal ratio of the parasitic capacitance may be controlled by adjusting the thickness of the buffer layer **102** between the dummy plate **210** and the bit line **110**.

[0121] FIGS. **6A** to **6C** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method of fabricating the semiconductor device of FIGS. **6A** to **6C** may be similar to that of FIGS. **2A** to **2K**. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. **2A** to **2K**, descriptions of FIGS. **2A** to **2K** may be referred to.

[0122] Referring to FIG. **6A**, a buffer layer **12** may be formed over the substrate **11**. A conductive layer **13A**, a barrier material layer **14A**, a lower interface material layer **15A**, a channel material layer **16A**, and an upper interface material layer **17A** may be sequentially formed over the buffer layer **12**.

[0123] Subsequently, a first contact layer **27A** and a second contact layer **28A** may be stacked over the upper interface material layer **17A**. The first contact layer **27A** may include titanium nitride, and the second contact layer **28A** may include tungsten.

[0124] Referring to FIG. **6B**, line structures **19L** may be formed. Each of the line structures **19L** may include a stack of a bit line **13**, a bit line barrier layer **14**, a lower interface layer **15B**, a channel material **16B**, an upper interface layer **17B**, a first contact layer **27B**, and the second contact layer **28B**. First trenches **19T** may be formed between the line structures **19L**.

[0125] Referring to FIG. **6C**, a portion of the line structures **19L** may be selectively etched. As a result, oxide semiconductor pillars **21P** may be formed over the bit line barrier layer **14**. Each of the oxide semiconductor pillars **21P** may include a stack of a lower interface layer **15**, a channel layer **16**, and an upper interface layer **17**. The lower interface layer **15**, the channel layer **16**, and the upper interface layer **17** may be formed by etching the lower interface layer **15B**, the channel material **16B**, and the upper interface layer **17B**, respectively. A stack of first contact plugs **27** and second contact plugs **28** may be formed over the upper interface layers **17**. The first contact plugs **27** may be formed by etching the first contact layer **27B**, and the second contact plugs **28** may be formed by etching the second contact layer **28B**.

[0126] Trenches **22** may be formed between the oxide semiconductor pillars **21P**. The first contact plugs **27** and the second contact plugs **28** may be cut by the trenches **22**, and accordingly, each of the first contact plugs **27** and the second contact plugs **28** may have a pillar shape.

[0127] Subsequently, a gate dielectric layer **23** may be formed on the sides of the oxide semiconductor pillars **21P**. The gate dielectric layer **23** may cover the sidewalls of the first contact plugs **27** and the second contact plugs **28**.

[0128] Subsequently, tapered vertical word lines **24** may be formed over the gate dielectric layer **23**.

[0129] Subsequently, as illustrated in FIG. **2K**, a capacitor **30** may be formed over the second contact plugs **28**.

[0130] FIGS. **7A** to **7G** are cross-sectional views illustrating a method for fabricating a

semiconductor device in accordance with another embodiment of the present disclosure. The method of fabricating the semiconductor device of FIGS. 7A to 7G may be similar to that of FIGS. 2A to 2K. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. 2A to 2K, descriptions of FIGS. 2A to 2K may be referred to.

[0131] Referring to FIG. 7A, a buffer layer **12** may be formed over the substrate **11**. A conductive layer **13A** and a barrier material layer **14A** may be formed over the buffer layer **12**.

[0132] Subsequently, a first sacrificial layer **15D** and a second sacrificial layer **18D** may be sequentially formed over the barrier material layer **14A**. The first sacrificial layer **15D** may include polysilicon, and the second sacrificial layer **18D** may include silicon nitride.

[0133] Referring to FIG. 7B, second sacrificial layer lines **18L** may be formed. The second sacrificial layer lines **18L** may be formed by etching the second sacrificial layer **18D**. The second sacrificial layer lines **18L** may be formed by a double patterning process.

[0134] Subsequently, a first sacrificial layer **15D** may be etched by using the second sacrificial layer lines **18L** as an etch barrier, and the barrier material layer **14A** and the conductive layer **13A** may be etched continuously.

[0135] A plurality of line structures **19L** may be formed over the buffer layer **12** by a series of the etching processes described above. Each of the line structures **19L** may include a stack of a bit line **13**, a bit line barrier layer **14**, a first sacrificial layer line **15L**, and a second sacrificial layer line **18L**. The bit line barrier layer **14** may be formed by etching the barrier material layer **14A**, and the bit line **13** may be formed by etching the conductive layer **13A**. The first sacrificial layer line **15L** may be formed by etching the first sacrificial layer **15D**. First trenches **19T** may be formed between the line structures **19L**.

[0136] Referring to FIG. 7C, dielectric lines **20** may be formed between the line structures **19L**. The dielectric lines **20** may include a dielectric material. For example, the dielectric lines **20** may include silicon oxide, silicon nitride, silicon carbon oxide (SiCO), a spin-on-dielectric layer, or a combination thereof. The dielectric lines **20** may be formed by depositing a dielectric material to fill the first trenches **19T** between the line structures **19L** and performing a planarization process of a dielectric material. The dielectric lines **20** may be formed by sequentially forming a silicon nitride, a silicon carbon oxide, and a spin-on dielectric layer and performing a planarization process.

[0137] Referring to FIG. 7D, a portion of the line structures **19L** may be selectively etched to form sacrificial pillars **21P'**. The bit line barrier layer **14** and the bit line **13** may be disposed below the sacrificial pillars **21P'**.

[0138] Each of the sacrificial pillars **21P'** may include a stack of a first sacrificial pillar **15LP** and a second sacrificial pillar **18LP**. The first sacrificial pillar **15LP** may be formed by etching the first sacrificial layer line **15L**, and the second sacrificial pillar **18LP** may be formed by etching the second sacrificial layer line **18L**.

[0139] Second trenches **22** may be formed between the sacrificial pillars **21P'**.

[0140] Subsequently, the gate dielectric layer **23** and the tapered vertical word line **24** may be formed by a series of processes as illustrated in FIGS. 2E and 2F. Referring to FIG. 7E, the gate dielectric layer **23** may be formed on the sides of the sacrificial pillars **21P'**. The tapered vertical word line **24** may be formed over the gate dielectric layer **23**. The bottom portion of the tapered vertical word line **24** may have a thinner thickness than the other portions.

[0141] Referring to FIG. 7F, an inter-layer dielectric layer **26** may be formed over the tapered vertical word line **24**. The inter-layer dielectric layer **26** may be planarized to expose the upper surfaces of the second sacrificial pillars **18LP**.

[0142] Subsequently, the second sacrificial pillars **18LP** and the first sacrificial pillars **15LP** may be selectively removed. Accordingly, pillar-shaped openings **18H** may be formed. The bottom surface of the pillar-shaped openings **18H** may expose the surface of the barrier layer **14**.

[0143] Referring to FIG. 7G, oxide semiconductor pillars **21P** filling the pillar-shaped openings

18H may be formed. Each of the oxide semiconductor pillars **21P** may include a stack of the lower interface layer **15**, the channel layer **16**, and the upper interface layer **17**. The lower interface layer **15**, the channel layer **16**, and the upper interface layer **17** may be formed by epitaxial growth, individually.

[0144] The lower interface layer **15** may contain indium. The lower interface layer **15** may include an indium-rich oxide semiconductor material. For example, the lower interface layer **15** may include indium-rich IGZO. The lower interface layer **15** may be formed to have a thickness of approximately 10 to 50 Å.

[0145] A channel layer **16** may be formed on the lower interface layer **15**. The channel layer **16** may contain indium. The channel layer **16** may include IGZO. The channel layer **16** may be formed to have a thickness of 200 to 1000 Å.

[0146] An upper interface layer **17** may be formed over the channel layer **16**. The upper interface layer **17** may contain indium. The upper interface layer **17** may include an indium-rich oxide semiconductor material. For example, the upper interface layer **17** may include indium-rich IGZO. The upper interface layer **17** may be formed to have a thickness of approximately 10 to 50 Å.

[0147] The oxide semiconductor pillars **21P** may partially fill the pillar-shaped openings **18H**, and thus hole-shaped recesses **18R** may be defined by the oxide semiconductor pillars **21P**.

[0148] Subsequently, a series of the processes as described in FIGS. **2I** to **2K** may be performed.

[0149] FIGS. **8** to **10** are cross-sectional views illustrating semiconductor devices in accordance with other embodiments of the present disclosure. FIGS. **8** to **10** may be similar to the semiconductor device **100** shown in FIGS. **1A** to **1E**, and also may be similar to the semiconductor device **300** shown in FIG. **5**. Hereinafter, as for the detailed descriptions on the constituent elements also appearing in FIGS. **1A** to **1E** and FIG. **5**, descriptions of FIGS. **1A** to **1E** and FIG. **5** may be referred to.

[0150] Referring to FIG. **8**, the semiconductor device **400** may include a substrate **101**, a buffer layer **102**, a bit line **110**, a barrier layer **111**, an oxide semiconductor pillar **120**, a tapered vertical word line **124**, a gate dielectric layer **125**, and a capacitor **130**. The semiconductor device **400** may further include a first contact plug **126** and a second contact plug **127** between the oxide semiconductor pillar **120** and the capacitor **130**. The oxide semiconductor pillar **120** may include a lower interface layer **122**, an oxide semiconductor channel layer **121**, and an upper interface layer **123**. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include an oxide semiconductor material. The lower interface layer **122**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include IGZO, but the lower interface layer **122** and the upper interface layer **123** may have a higher indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may be IGZO, and the lower interface layer **122** and the upper interface layer **123** may be indium-rich IGZO.

[0151] The semiconductor device **400** may further include a dummy plate **210** below the bit line **110**. The buffer layer **102** may be disposed between the dummy plate **210** and the bit line **110**.

According to another embodiment of the present disclosure, the dummy plate **210** may be omitted.

[0152] The semiconductor device **400** may further include an isolating dielectric layer **401** which is disposed between the tapered vertical word line **124** and the barrier layer **111**. The isolating dielectric layer **401** may be disposed below the gate dielectric layer **125**. The isolating dielectric layer **401** may further increase the distance between the lower level portion **124L** of the tapered vertical word line **124** and the bit line **110** (refer to a reference numeral **H1**).

[0153] Referring to FIG. **9**, the semiconductor device **410** may include a substrate **101**, a buffer layer **102**, a bit line **110**, a barrier layer **111**, an oxide semiconductor pillar **120**, a tapered vertical word line **124**, a gate dielectric layer **125**, and a capacitor **130**. The semiconductor device **410** may further include a first contact plug **126** and a second contact plug **127** between the oxide semiconductor pillar **120** and the capacitor **130**. The oxide semiconductor pillar **120** may include a

lower interface layer **122'**, an oxide semiconductor channel layer **121**, and an upper interface layer **123**. The lower interface layer **122'**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include IGZO, but the lower interface layer **122'** and the upper interface layer **123** may have a higher indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may be IGZO, and the lower interface layer **122'** and the upper interface layer **123** may be indium-rich IGZO.

[0154] The semiconductor device **410** may further include a dummy plate **210** below the bit line **110**. The buffer layer **102** may be disposed between the dummy plate **210** and the bit line **110**.

According to another embodiment of the present disclosure, the dummy plate **210** may be omitted.

[0155] The semiconductor device **410** may further include an isolating dielectric layer **401** which is disposed between the tapered vertical word line **124** and the barrier layer **111**. The isolating dielectric layer **401** may further increase the distance between the lower level portion **124L** of the tapered vertical word line **124** and the bit line **110** (refer to a reference numeral H1).

[0156] The height of the lower interface layer **122'** of the oxide semiconductor pillar **120** may be higher than the height of the upper interface layer **123**. Also, the height of the lower interface layer **122'** may be higher than the height of the lower interface layer **122** of FIG. 8. The lower interface layer **122'** and the isolating dielectric layer **401** may have the same height.

[0157] Referring to FIG. 10, the semiconductor device **420** may include a substrate **101**, a buffer layer **102**, a bit line **110**, a barrier layer **111**, an oxide semiconductor pillar **120**, a tapered vertical word line **124**, a gate dielectric layer **125**, and a capacitor **130**. The semiconductor device **420** may further include a first contact plug **126** and a second contact plug **127** between the oxide semiconductor pillar **120** and the capacitor **130**. The oxide semiconductor pillar **120** may include a lower interface layer **122''**, an oxide semiconductor channel layer **121**, and an upper interface layer **123**. The lower interface layer **122''**, the oxide semiconductor channel layer **121**, and the upper interface layer **123** may all include IGZO, but the lower interface layer **122''** and the upper interface layer **123** may have a higher indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may be IGZO, and the lower interface layer **122''** and the upper interface layer **123** may be indium-rich IGZO.

[0158] The semiconductor device **420** may further include a dummy plate **210** below the bit line **110**. The buffer layer **102** may be disposed between the dummy plate **210** and the bit line **110**.

According to another embodiment of the present disclosure, the dummy plate **210** may be omitted.

[0159] The semiconductor device **420** may further include an isolating dielectric layer **401** which is disposed between the tapered vertical word line **124** and the gate dielectric layer **125**. The isolating dielectric layer **401** may be disposed over the gate dielectric layer **125**. The isolating dielectric layer **401** may further increase the distance between the lower level portion **124L** of the tapered vertical word line **124** and the bit line **110** (refer to a reference numeral H1).

[0160] The height of the lower interface layer **122''** of the oxide semiconductor pillar **120** may be greater than the height of the upper interface layer **123**. The height of the lower interface layer **122''** may be higher than the height of the lower interface layer **122'** of FIG. 9. According to another embodiment of the present disclosure, the height of the lower interface layer **122''** may be the same as the height of the lower interface layer **122'** of FIG. 9. According to another embodiment of the present disclosure, the height of the lower interface layer **122''** may be the same as the height of the lower interface layer **122** of FIG. 8.

[0161] In the semiconductor devices **400**, **410**, and **420** shown in FIGS. 8 to 10, the tapered vertical word line **124** and the bit line **110** may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer **401** and the gate dielectric layer **125** (refer to a reference numeral H1). The isolating dielectric layer **401** may include silicon oxide.

[0162] The gate dielectric layer **125** of the semiconductor devices **400**, **410**, and **420** shown in FIGS. 8 to 10 may include a vertical portion **125V** which is disposed between the oxide semiconductor pillar **120** and the tapered vertical word line **124**, and a horizontal portion **125F**

which extends from the vertical portion **125V** and is disposed between the bit line **110** and the tapered vertical word line **124**. The isolating dielectric layer **401** may be disposed at a lower level than the horizontal portion **125F** of the gate dielectric layer **125** (refer to FIGS. **8** and **9**) or may be disposed at a higher level than the horizontal portion **125F** of the gate dielectric layer **125** (refer to FIG. **10**).

[0163] FIGS. **11A** to **14B** are embodiments of a method for fabricating the semiconductor devices in accordance with FIGS. **8** to **10**.

[0164] FIGS. **11A** to **11C** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. **11A** to **11C** may be similar to that of FIGS. **2A** to **2K**. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. **2A** to **2K**, descriptions of FIGS. **2A** to **2K** may be referred to. FIGS. **11A** to **11C** illustrate the fabrication method according to the line B-B' shown in FIG. **1A**.

[0165] First, referring to FIGS. **2A** to **2D**, a buffer layer **12** may be formed over the substrate **11**, and a bit line **13** and a barrier layer **14** may be formed over the buffer layer **12**. Subsequently, oxide semiconductor pillars **21P** including a lower interface layer **15**, a channel layer **16**, and an upper interface layer **17** may be formed over the barrier layer **14**. Sacrificial pillars **18P** may be disposed over the upper interface layer **17**. Second trenches **22** may be formed between the oxide semiconductor pillars **21P**.

[0166] Subsequently, referring to FIG. **11A**, an isolating dielectric layer **31** may be formed over the barrier layer **14**. The isolating dielectric layer **31** may be formed by depositing a dielectric layer **31A** over the oxide semiconductor pillars **21P** and then performing an etch-back process onto the dielectric layer **31A**. The isolating dielectric layer **31** may include silicon oxide. The height of the isolating dielectric layer **31** may be higher than the height of the lower interface layer **15**.

[0167] Referring to FIG. **11B**, a gate dielectric layer **23** may be formed over the isolating dielectric layer **31**. The gate dielectric layer **23** may be formed on the exposed sidewalls of the oxide semiconductor pillars **21P** and the sacrificial pillars **18P**. The isolating dielectric layer **31** may be disposed between the gate dielectric layer **23** and the bit line **13**. The isolating dielectric layer **31** may be disposed at a lower level than the gate dielectric layer **23**.

[0168] Referring to FIG. **11C**, the tapered vertical word lines **24** may be formed. Tapered vertical word lines **24** may be formed with reference to the methods illustrated in FIGS. **2E** to **2G**. The gate dielectric layer **23**, the isolating dielectric layer **31**, and the barrier layer **14** may be disposed between the bottom portion of the tapered vertical word lines **24** and the bit line **13**. The tapered vertical word lines **24** and the bit line **13** may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer **31** and the gate dielectric layer **23** (refer to a reference numeral **H1**).

[0169] FIGS. **12A** and **12B** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. **12A** and **12B** may be similar to that of FIGS. **2A** to **2K**. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. **2A** to **2K**, descriptions of FIGS. **2A** to **2K** may be referred to. FIGS. **12A** and **12B** may be the fabrication method according to the line B-B' shown in FIG. **1A**.

[0170] First, referring to FIGS. **2A** to **2D**, a buffer layer **12** may be formed over the substrate **11**, and a bit line **13** and a barrier layer **14** may be formed over the buffer layer **12**. Subsequently, oxide semiconductor pillars **21P** including a lower interface layer **15**, a channel layer **16**, and an upper interface layer **17** may be formed over the barrier layer **14**. Sacrificial pillars **18P** may be disposed over the upper interface layer **17**.

[0171] Subsequently, referring to FIG. **12A**, a gate dielectric layer **23** may be formed over the barrier layer **14**. The gate dielectric layer **23** may be formed on the exposed sidewalls of the oxide semiconductor pillars **21P** and the sacrificial pillars **18P**.

[0172] Subsequently, an isolating dielectric layer **31** may be formed over the gate dielectric layer **23**. The isolating dielectric layer **31** may be formed by depositing a dielectric layer **31A** over the gate dielectric layer **23** and performing an etch-back process onto the dielectric layer **31A**. The isolating dielectric layer **31** may include silicon oxide. The height of the isolating dielectric layer **31** may be higher than the height of the lower interface layer **15**. A gate dielectric layer **23** may be disposed between the isolating dielectric layer **31** and the bit line **13**.

[0173] As described above, the isolating dielectric layer **31** may be formed after the gate dielectric layer **23** is formed.

[0174] Referring to FIG. **12B**, tapered vertical word lines **24** may be formed. The tapered vertical word lines **24** may be formed with reference to the methods illustrated in FIGS. **2E** to **2G**. A gate dielectric layer **23**, an isolating dielectric layer **31**, and a barrier layer **14** may be disposed between the bottom portion of the tapered vertical word lines **24** and the bit line **13**. The tapered vertical word lines **24** and the bit line **13** may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer **31** and the gate dielectric layer **23** (refer to a reference numeral **H1**).

[0175] FIGS. **13A** to **13D** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. **13A** to **13D** may be similar to that of FIGS. **2A** to **2K**. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. **2A** to **2K**, descriptions of FIGS. **2A** to **2K** may be referred to. FIGS. **13A** to **13D** may be the fabrication method according to the line B-B' shown in FIG. **1A**.

[0176] Referring to FIGS. **2A** and **13A**, a buffer layer **12** may be formed over a substrate **11**, and a conductive layer **13A** and a barrier material layer **14A** may be sequentially formed over the buffer layer **12**.

[0177] A lower interface layer **15C** may be formed over the barrier material layer **14A**. The lower interface layer **15C** may include a conductive material. The lower interface layer **15C** may include an oxide semiconductor material. The lower interface layer **15C** may contain indium. The lower interface layer **15C** may include an indium-rich oxide semiconductor material. For example, the lower interface layer **15C** may include indium-rich IGZO. The lower interface layer **15C** may be formed to have a thickness of approximately 10 to 50 Å. The lower interface layer **15C** may be thicker than the lower interface material layer **15A** shown in FIG. **2A**.

[0178] A channel material layer **16A** may be formed over the lower interface layer **15C**. The channel material layer **16A** may include a conductive material. The channel material layer **16A** may include an oxide semiconductor material. The channel material layer **16A** may contain indium. The channel material layer **16A** may include IGZO. The channel material layer **16A** may be formed to have a thickness of approximately 200 to 1000 Å.

[0179] An upper interface material layer **17A** may be formed over the channel material layer **16A**. The upper interface material layer **17A** may include a conductive material. The upper interface material layer **17A** may include an oxide semiconductor material. The upper interface material layer **17A** may contain indium. The upper interface material layer **17A** may include an indium-rich oxide semiconductor material. For example, the upper interface material layer **17A** may include indium-rich IGZO. The upper interface material layer **17A** may be formed to have a thickness of approximately 10 to 50 Å. The upper interface material layer **17A** may be thinner than the lower interface layer **15C**.

[0180] As described above, the lower interface layer **15C**, the channel material layer **16A**, and the upper interface material layer **17A** may be vertically stacked over the barrier material layer **14A**. The lower interface layer **15C**, the channel material layer **16A**, and the upper interface material layer **17A** may all include an oxide semiconductor material. The lower interface layer **15C**, the channel material layer **16A**, and the upper interface material layer **17A** may all include IGZO, but the lower interface layer **15C** and the upper interface material layer **17A** may have a higher indium concentration than the channel material layer **16A**. The channel material layer **16A** may be IGZO,

and the lower interface material layer **15A** and the upper interface material layer **17A** may be indium-rich IGZO. Since the lower interface layer **15C** and the upper interface material layer **17A** contain a high concentration of indium, the resistance may be reduced lower than that of the channel material layer **16A**. Also, channel seamless interconnection of the channel material layer **16A** is possible.

[0181] Subsequently, a sacrificial layer **18A** may be formed over the upper interface material layer **17A**. The sacrificial layer **18A** may include a stack of different materials. The sacrificial layer **18A** may include silicon nitride.

[0182] Referring to FIG. **13B**, a bit line **13** and a barrier layer **14** may be formed over the buffer layer **12** by a series of the processes as illustrated in FIGS. **2B** to **2D**. Subsequently, oxide semiconductor pillars **21P** including a lower interface layer **15'**, a channel layer **16**, and an upper interface layer **17** may be formed over the barrier layer **14**. Sacrificial pillars **18P** may be disposed over the upper interface layer **17**. Second trenches **22** may be formed between the oxide semiconductor pillars **21P**.

[0183] Referring to FIG. **13C**, an isolating dielectric layer **31** may be formed over the barrier layer **14**. A method of forming the isolating dielectric layer **31** may be the same as that of FIG. **11A**. The height of the isolating dielectric layer **31** and the lower interface layer **15'** may be the same as each other.

[0184] Referring to FIG. **13D**, a gate dielectric layer **23** may be formed over the isolating dielectric layer **31**, and tapered vertical word lines **24** may be formed over the gate dielectric layer **23**. The tapered vertical word lines **24** may be formed with reference to the methods illustrated in FIGS. **2E** to **2G**. A gate dielectric layer **23**, an isolating dielectric layer **31**, and a barrier layer **14** may be disposed between the bottom portion of the tapered vertical word lines **24** and the bit line **13**. The tapered vertical word lines **24** and the bit line **13** may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer **31** and the gate dielectric layer **23** (refer to a reference numeral **H1**).

[0185] FIGS. **14A** and **14B** are cross-sectional views illustrating a method for fabricating a semiconductor device in accordance with another embodiment of the present disclosure. The method for fabricating the semiconductor device of FIGS. **14A** and **14B** may be similar to that of FIGS. **2A** through **2K**. Hereinafter, for the detailed descriptions of the constituent elements also appearing in FIGS. **2A** to **2K**, descriptions of FIGS. **2A** to **2K** may be referred to. FIGS. **14A** and **14B** may be the fabrication method according to the line B-B' shown in FIG. **1A**.

[0186] First, as illustrated in FIGS. **2A** to **2D** and **13A** and **13B**, a buffer layer **12** may be formed over the substrate **11**, and a bit line **13** and a barrier layer **14** may be formed over the buffer layer **12**. Subsequently, oxide semiconductor pillars **21P** including a lower interface layer **15'**, a channel layer **16**, and an upper interface layer **17** may be formed over the barrier layer **14**. Sacrificial pillars **18P** may be disposed over the upper interface layer **17**.

[0187] Subsequently, referring to FIG. **14A**, a gate dielectric layer **23** may be formed over the barrier layer **14**. The gate dielectric layer **23** may be formed on the exposed sidewalls of the oxide semiconductor pillars **21P** and the sacrificial pillars **18P**.

[0188] Subsequently, an isolating dielectric layer **31** may be formed over the gate dielectric layer **23**. The isolating dielectric layer **31** may be formed by depositing the dielectric layer **31A** over the gate dielectric layer **23** and performing an etch-back process onto the dielectric layer **31A**. The isolating dielectric layer **31** may include silicon oxide. The gate dielectric layer **23** may be disposed between the isolating dielectric layer **31** and the bit line **13**.

[0189] As described above, the isolating dielectric layer **31** may be formed after the gate dielectric layer **23** is formed.

[0190] Referring to FIG. **14B**, tapered vertical word lines **24** may be formed. The tapered vertical word lines **24** may be formed with reference to the methods shown in FIGS. **2E** to **2G**. The gate dielectric layer **23**, the isolating dielectric layer **31**, and the barrier layer **14** may be disposed

between the bottom portion of the tapered vertical word lines **24** and the bit line **13**. The tapered vertical word lines **24** and the bit line **13** may be spaced apart from each other by a sufficient distance due to the isolating dielectric layer **31** and the gate dielectric layer **23** (refer to a reference numeral **H1**).

[0191] FIG. **15A** is a schematic plan view illustrating a semiconductor device **500** in accordance with an embodiment of the present disclosure. FIG. **15B** is a cross-sectional view taken along a line A-A' shown in FIG. **15A**. FIG. **15C** is a cross-sectional view taken along a line B-B' shown in FIG. **15A**. FIG. **15D** is an enlarged view illustrating a portion **501** of FIG. **15A**. The semiconductor device **500** of FIGS. **15A** to **15D** may be similar to the semiconductor device **100** of FIGS. **1A** to **1E**. Hereinafter, for the detailed description of the constituent elements of the semiconductor device **500** also appearing in FIGS. **1A** to **1E**, FIGS. **1A** to **1E** may be referred to.

[0192] Referring to FIGS. **15A** to **15D**, the semiconductor device **500** may include a lower horizontal conductive line, an upper horizontal conductive line, a shield line, a vertically oriented pillar, and a data storage element. The lower horizontal conductive line may include a bit line **110**. The upper horizontal conductive line may include a main gate **124S**. The vertically oriented pillar may include an oxide semiconductor pillar **120**. The data storage element may include a capacitor **130**. The shield line may include a shield gate **124G**.

[0193] The semiconductor device **500** may include the bit line **110**, the oxide semiconductor pillar **120** extending vertically from the bit line **110**, the capacitor **130** over the oxide semiconductor pillar **120**, and the main gate **124S** disposed on a sidewall of the oxide semiconductor pillar **120**.

[0194] Referring back to FIG. **15B**, the oxide semiconductor pillar **120** may include a first oxide semiconductor interface layer **122** coupled to the bit line **110**, a second oxide semiconductor interface layer **123** coupled to the capacitor **130**, and an oxide semiconductor channel layer **121** between the first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123**. Each of the first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123** may include an oxide semiconductor layer. The first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123** may be referred to as a lower oxide semiconductor interface layer and an upper oxide semiconductor interface layer, respectively. The semiconductor device **500** may include a dynamic random access memory (DRAM). As shown in FIG. **15A**, the bit line **110** may extend in the first direction **D1**, and the main gate **124S** may extend in the second direction **D2**. The first direction **D1** and the second direction **D2** may intersect with each other perpendicularly.

[0195] The bit line **110** may be disposed at a lower level than the oxide semiconductor channel layer **121**. The capacitor **130** may be disposed at a higher level than the oxide semiconductor channel layer **121**. As shown in FIG. **15C**, the main gate **124S** may be disposed on the sidewall of a first side of the oxide semiconductor channel layer **121**. The first oxide semiconductor interface layer **122** may be disposed between the bit line **110** and the oxide semiconductor channel layer **121**. The second oxide semiconductor interface layer **123** may be disposed between the capacitor **130** and the oxide semiconductor channel layer **121**. The bit line **110**, the first oxide semiconductor interface layer **122**, the oxide semiconductor channel layer **121**, the second oxide semiconductor interface layer **123**, and the capacitor **130** may be disposed vertically in the third direction **D3**.

[0196] The semiconductor device **500** may be described in detail as follows.

[0197] Referring back to FIGS. **15A**, **15B** and **15C**, a buffer layer **102** may be formed over a substrate **101**, and the bit line **110** may be formed over the buffer layer **102**. A vertical channel transistor TR may be formed over the bit line **110**, and the capacitor **130** may be formed over the vertical channel transistor TR. The vertical channel transistor TR may include the oxide semiconductor pillar **120**, the main gate **124S** disposed on the sidewall of the first side of the oxide semiconductor pillar **120**, and a gate dielectric layer **125** between the oxide semiconductor pillar **120** and the main gate **124S**. The oxide semiconductor pillar **120** may include the oxide semiconductor channel layer **121**, the first oxide semiconductor interface layer **122** between the

oxide semiconductor channel layer **121** and the bit line **110**, and the second oxide semiconductor interface layer **123** between the oxide semiconductor channel layer **121** and the capacitor **130**. A barrier layer **111** may be disposed between the bit line **110** and the first oxide semiconductor interface layer **122**.

[0198] One bit line **110**, one vertical channel transistor TR, and one capacitor **130** may form a memory cell MC. The memory cell MC may have an one transistor, one capacitor (1T-1C or 1T1C) structure. The semiconductor device **500** may include a memory cell array MCA, that is, an array of memory cells MC. The memory cell array MCA may include a plurality of first arrays AR1 and a plurality of second arrays AR21 to AR24. The first arrays AR1 may be disposed spaced apart from each other in the second direction D2. The second arrays AR21 to AR24 may be disposed spaced apart from each other in the first direction D1. The first arrays AR1 and the second arrays AR21 to AR24 may intersect with each other. The first arrays AR1 may be column arrays, while the second arrays AR21 to AR24 may be row arrays. The first arrays AR1 may include a plurality of memory cells MC that are disposed in the first direction D1. The second arrays AR21 to AR24 may include a plurality of memory cells MC that are disposed in the second direction D2. In each first array AR1, a plurality of memory cells MC may share one bit line **110**. In each second array AR21 to AR24, a plurality of memory cells MC may share one main gate **124S**. The second arrays AR21 to AR24 may include a first sub-array AR21, a second sub-array AR22, a third sub-array AR23, and a fourth sub-array AR24. A first space S1 between the first sub-array AR21 and the second sub-array AR22 may be the same as a second space S2 between the third sub-array AR23 and the fourth sub-array AR24. A third space S3 between the second sub-array AR22 and the third sub-array AR23 may be greater than each of the first and second spaces S1 and S2. According to another embodiment of the present disclosure, the number of the sub-arrays may be four or more. The first and second spaces S1 and S2 may be referred to as shield trenches.

[0199] The semiconductor device **500** may further include shield gates **124G**. The shield gates **124G** may be disposed between the main gates **124S**. The shield gates **124G** may be disposed between the neighboring vertical channel transistors TR. The shield gates **124G** may be disposed between the neighboring memory cells MC.

[0200] Referring back to FIG. 15A, the shield gates **124G** may be disposed between the neighboring second arrays AR21 to AR24. The shield gates **124G** may include a first shield gate **124G1** and a second shield gate **124G2**.

[0201] For example, the first shield gate **124G1** may be disposed between the first sub-array AR21 and the second sub-array AR22. To be specific, the first shield gate **124G1** may be disposed between the vertical channel transistors TR of the first sub-array AR21 and the vertical channel transistors TR of the second sub-array AR22. Since the first shield gate **124G1** is disposed between the first sub-array AR21 and the second sub-array AR22, the interference between the vertical channel transistors TR of the first sub-array AR21 and the vertical channel transistors TR of the second sub-array AR22 may be blocked off. The first sub-array AR21 and the second sub-array AR22 may share the first shield gate **124G1**.

[0202] The second shield gate **124G2** may be disposed between the third sub-array AR23 and the fourth sub-array AR24. To be specific, the second shield gate **124G2** may be disposed between the vertical channel transistors TR of the third sub-array AR23 and the vertical channel transistors TR of the fourth sub-array AR24. Since the second shield gate **124G2** is disposed between the third sub-array AR23 and the fourth sub-array AR24, the interference between the vertical channel transistors TR of the third sub-array AR23 and the vertical channel transistors TR of the fourth sub-array AR24 may be blocked off. The third sub-array AR23 and the fourth sub-array AR24 may share the second shield gate **124G2**.

[0203] Referring back to FIGS. 15B and 15C, the semiconductor device **500** may further include first and second contact plugs **126** and **127** between the second oxide semiconductor interface layer **123** and the capacitor **130**.

[0204] The substrate **101** may be a material appropriate for semiconductor processing. The substrate **101** may include a semiconductor substrate. The substrate **101** may be formed of a silicon-containing material. The substrate **101** may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate **101** may also include another semiconductor material, such as germanium. The substrate **101** may include a III/V-group semiconductor substrate, for example, a compound semiconductor substrate, such as gallium arsenide (GaAs). The substrate **101** may also include a Silicon-On-Insulator (SOI) substrate.

[0205] The buffer layer **102** may include silicon oxide, silicon nitride, or a combination thereof. The buffer layer **102** may be formed of silicon oxide to reduce parasitic capacitance. For example, the buffer layer **102** may include Tetraethyl Orthosilicate (TEOS).

[0206] The bit line **110** may extend in the first direction D1 over the buffer layer **102**. The bit line **110** may include a metal-based material. The bit line **110** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The bit line **110** may have a thickness of approximately 100 to 400 Å. The bit line **110** may include a tungsten layer.

[0207] The barrier layer **111** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier layer **111** may include titanium nitride, molybdenum, or ruthenium. The barrier layer **111** may have a thickness of approximately 10 to 50 Å. For example, the barrier layer **111** may include a titanium nitride layer.

[0208] Each of the first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123** may include an oxide semiconductor material having a lower resistance than the oxide semiconductor channel layer **121**. Each of the first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123** may include a metal-rich oxide semiconductor material. The oxide semiconductor channel layer **121** may include an oxygen-rich oxide semiconductor material. For example, the oxide semiconductor channel layer **121** may include indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO) or zinc tin oxide (ZTO), and each of the first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123** may include indium-rich IGZO. Indium-rich IGZO may refer to IGZO whose indium content is greater than gallium (Ga) and zinc (Zinc). For example, the content of indium may be approximately 40% or more.

[0209] The oxide semiconductor channel layer **121** may include an oxide semiconductor material. The oxide semiconductor channel layer **121** may contain indium. The oxide semiconductor channel layer **121** may include IGZO. The oxide semiconductor channel layer **121** may be formed to have a thickness of approximately 200 to 1,000 Å.

[0210] The first oxide semiconductor interface layer **122** may include an oxide semiconductor material. The first oxide semiconductor interface layer **122** may contain indium. The first oxide semiconductor interface layer **122** may include an indium-rich oxide semiconductor material. For example, the first oxide semiconductor interface layer **122** may include indium-rich IGZO. The first oxide semiconductor interface layer **122** may be formed to have a thickness of approximately 10 to 50 Å.

[0211] The second oxide semiconductor interface layer **123** may include an oxide semiconductor material. The second oxide semiconductor interface layer **123** may contain indium. The second oxide semiconductor interface layer **123** may include an indium-rich oxide semiconductor material. For example, the second oxide semiconductor interface layer **123** may include indium-rich IGZO. The second oxide semiconductor interface layer **123** may be formed to have a thickness of approximately 10 to 50 Å.

[0212] As described above, the oxide semiconductor pillar **120** may extend vertically in the third direction D3 over the bit line **110**. The oxide semiconductor pillar **120** may include the first oxide semiconductor interface layer **122**, the oxide semiconductor channel layer **121**, and the second

oxide semiconductor interface layer **123** that are vertically stacked over the bit line **110** in the mentioned order. The first oxide semiconductor interface layer **122**, the oxide semiconductor channel layer **121**, and the second oxide semiconductor interface layer **123** may all include an oxide semiconductor material. The first oxide semiconductor interface layer **122**, the oxide semiconductor channel layer **121**, and the second oxide semiconductor interface layer **123** may all include IGZO (indium gallium zinc oxide). The first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123** may have a higher indium concentration than the oxide semiconductor channel layer **121**. The oxide semiconductor channel layer **121** may include IGZO, and each of the first oxide semiconductor interface layer **122** and the second oxide semiconductor interface layer **123** may include indium-rich IGZO. The first oxide semiconductor interface layer **122**, the oxide semiconductor channel layer **121**, and the second oxide semiconductor interface layer **123** may be referred to as an active pillar.

[0213] The main gate **124S** of a single structure may be disposed on a sidewall of the oxide semiconductor channel layer **121**. The main gate **124S** may be referred to as a tapered vertical word line. The main gate **124S** and the bit line **110** may extend in the directions intersecting with each other. The main gate **124S** may have a reverse tapered shape in the third direction **D3**. The reverse tapered shape may refer to a shape whose bottom portion has a narrower width than the upper portion. For example, the main gate **124S** may include a lower level portion **124L** disposed adjacent to the bit line **110** and an upper level portion **124U** disposed adjacent to the capacitor **130**. The thickness of the lower level portion **124L** in the first direction **D1** may be thinner than the thickness of the upper level portion **124U**. The lower level portion **124L** may be disposed adjacent to the first oxide semiconductor interface layer **122**. The upper level portion **124U** may be disposed adjacent to the second oxide semiconductor interface layer **123**. The lower level portion **124L** and the upper level portion **124U** may be formed of the same material. The lower level portion **124L** and the bit line **110** may be disposed spaced apart from each other with a gap. The lower level portion **124L** of the main gate **124S** may not contact the bit line **110** and the barrier layer **111**.

[0214] The main gate **124S** may include a metal-based material. The main gate **124S** may include a metal, a metal nitride, or a combination thereof. The main gate **124S** may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0215] The gate dielectric layer **125** may be formed between the main gate **124S** and the oxide semiconductor pillar **120**. The gate dielectric layer **125** may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The gate dielectric layer **125** may be disposed between the oxide semiconductor pillar **120** and the main gate **124S**. The gate dielectric layer **125** may include a horizontal portion that extends to be disposed between the lower level portion **124L** of the main gate **124S** and the bit line **110**, and the horizontal portion of the gate dielectric layer **125** may directly contact the barrier layer **111**. The oxide semiconductor pillar **120**, the gate dielectric layer **125**, and the main gate **124S** may form a vertical channel transistor TR. The main gate **124S** may be referred to as a tapered vertical gate.

[0216] The shield gate **124G** may be disposed between the neighboring oxide semiconductor pillars **120**. The shield gate **124G** may have the same width at the top and bottom portions. The gate dielectric layer **125** may be disposed between the shield gate **124G** and the oxide semiconductor pillars **120**. The shield gate **124G** may include a metal-based material. The shield gate **124G** may include a metal, a metal nitride, or a combination thereof. The main gate **124S** may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0217] The main gate **124S** may be referred to as a front gate, and the shield gate **124G** may be referred to as a common shield gate.

[0218] The main gate **124S** may function to control the vertical channel transistor TR, and the shield gate **124G** may function to block off the interference between the neighboring vertical channel transistors TR. The shield gate **124G** may be referred to as a back gate, a shield word line,

a common gate, or a shield line.

[0219] Referring back to FIGS. **15B** and **15C**, the contact plugs **126** and **127** may include a first contact plug **126** and a second contact plug **127** over the first contact plug **126**. The first contact plug **126** may directly contact the second oxide semiconductor interface layer **123**. The second contact plug **127** may directly contact the capacitor **130**. As shown in FIG. **15B**, the first contact plug **126** and the second contact plug **127** may vertically overlap with each other in the A-A' direction. As shown in FIG. **15C**, the first contact plug **126** and the second contact plug **127** may vertically partially overlap with each other in the B-B' direction. The first contact plug **126** and the second contact plug **127** may include a metal-based material. Each of the first contact plug **126** and the second contact plug **127** may include a metal, a metal nitride, or a combination thereof. The first contact plug **126** and the second contact plug **127** may be formed of the same metal-based material. According to another embodiment of the present disclosure, the first contact plug **126** and the second contact plug **127** may be formed of different metal-based materials. The sidewalls of the first contact plug **126** may be surrounded by the gate dielectric layer **125**.

[0220] Referring back to FIG. **1E**, the capacitor **130** may include the lower electrode **132**, the dielectric layer **135**, and the upper electrode **136**. The lower electrode **132** may be formed over the second contact plug **127**. The lower electrode **132** may have a pillar shape. The lower electrodes **132** may be supported by supporters **133** and **134**. The sidewalls of the bottom portions of the lower electrodes **132** may contact the etch stop layer **131**. According to another embodiment of the present disclosure, the lower electrode **132** may have a cylindrical shape.

[0221] In some embodiments, dielectric pillars may be disposed between the oxide semiconductor pillars **120** in the A-A' direction.

[0222] Referring to FIG. **15D**, two neighboring oxide semiconductor pillars **120A** and **120B** will be described as follows. From the perspective of a top view, each of the first and second oxide semiconductor pillars **120A** and **120B** may include an outer side **F1** and an inner side **F2** which faces the outer side **F1**. The outer sides **F1** may be disposed adjacent to first and second main gates **124S1** and **124S2**. The inner sides **F2** may be disposed adjacent to the shield gate **124G**. The first and second main gates **124S1** and **124S2** may have a tapered shape. Each of the first and second main gates **124S1** and **124S2** may include a first sidewall **VS1** having a straight profile, and a second sidewall **VS2** facing the first sidewall **VS1** and having a tapered profile. The second sidewall **VS2** may include a first segment **VS21** of a straight profile, a second segment **VS22** of a straight profile, and a third segment **VS23** of a tapered profile between the first segment **VS21** and the second segment **VS22**. In this way, the second sidewall **VS2** may have a straight profile and a tapered profile merged therein. The first sidewalls **VS1** of the first and second main gates **124S1** and **124S2** may be close to the outer sides **F1** of the first and second oxide semiconductor pillars **120A** and **120B**. The gate dielectric layer **125** may be disposed between the outer sides **F1** of the first and second oxide semiconductor pillars **120A** and **120B** and the first and second main gates **124S1** and **124S2**. A shield gate dielectric layer **125G** may be disposed between the shield gate **124G** and the inner sides **F2** of the first and second oxide semiconductor pillars **120A** and **120B**. The first and second oxide semiconductor pillars **120A** and **120B** may be disposed adjacent to each other with a shield trench **ST** therebetween. The shield trench **ST** may correspond to the first and second spaces **S1** and **S2** illustrated in FIG. **15A**. Each of the first and second oxide semiconductor pillars **120A** and **120B** may include an inner side **F2** that defines the inner sidewalls of the shield trench **ST** and an outer side **F1** that faces the inner side **F2**. The shield gate **124G** may be disposed between the first oxide semiconductor pillar **120A** and the second oxide semiconductor pillar **120B**. The first main gate **124S1** may be close to the outer side **F1** of the first oxide semiconductor pillar **120A**. The second main gate **124S2** may be close to the outer side **F1** of the second oxide semiconductor pillar **120B**. The shield gate dielectric layer **125G** may be referred to as a 'common gate dielectric layer', and the shield gate **124G** may be referred to as a 'common gate'.

[0223] Referring to FIGS. **15A** to **15D**, the semiconductor device **500** may include first and second

pillars **120** disposed adjacent to each other with a shield trench **ST** therebetween and each including an inner side **F2** that defines the inner sidewalls of the shield trench **ST** and an outer side **F1** that faces the inner side **F2**, the shield gate **124G** disposed in the shield trench **ST** between the first pillar **120** and the second pillar **120**, a first main gate **124S1** close to the outer side **F1** of the first pillar **120**, a second main gate **124S2** close to the outer side **F1** of the second pillar **120**, the bit line **110** disposed in the lower portions of the first and second pillars **120** and electrically connected to the first and second pillars **120**, and the capacitors **130** disposed in the upper portions of the first and second pillars **120** and electrically connected to the first and second pillars **120**, respectively. [0224] As described above, by forming the main gate **124S** and the shield gate **124G** as a single structure, the space between the neighboring vertical channel transistors **TR** may be reduced. Accordingly, since the space between the neighboring cells may be reduced, the cell area may be decreased, thereby increasing the degree of integration or securing the process margin.

[0225] FIGS. **16A** to **16I** illustrate a method for fabricating a semiconductor device in accordance with an embodiment of the present disclosure. FIGS. **16A** to **16I** are cross-sectional views simultaneously illustrating the fabrication method according to the A-A' and B-B' lines shown in FIG. **15A**.

[0226] Referring to FIG. **16A**, a buffer layer **12** may be formed over a substrate **11**. The substrate **11** may be a material appropriate for semiconductor processing. The substrate **11** may include a semiconductor substrate. The substrate **11** may be formed of a silicon-containing material. The substrate **11** may include silicon, monocrystalline silicon, polysilicon, amorphous silicon, silicon germanium, monocrystalline silicon germanium, polycrystalline silicon germanium, carbon-doped silicon, a combination thereof, or a multi-layer thereof. The substrate **11** may include another semiconductor material such as germanium. The substrate **11** may include a III/V-group semiconductor substrate, for example, a compound semiconductor substrate, such as gallium arsenide (GaAs). The substrate **11** may include a Silicon-On-Insulator (SOI) substrate. The buffer layer **12** may include a dielectric material. The buffer layer **12** may include silicon oxide, silicon nitride, or a combination thereof.

[0227] A conductive layer **13A** may be formed over the buffer layer **12**. The conductive layer **13A** may include a metal-based material. The conductive layer **13A** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The conductive layer **13A** may be formed by an Atomic Layer Deposition (ALD) process, a Physical Vapor Deposition (PVD) process, or a Chemical Vapor Deposition (CVD) process. The conductive layer **13A** may be formed to have a thickness of approximately 100 to 400 Å. For example, the conductive layer **13A** may be formed by depositing a tungsten layer to a thickness of approximately 200 Å through a Physical Vapor Deposition (PVD) process. The buffer layer **12** may be formed of silicon oxide to reduce the parasitic capacitance between the substrate **11** and the conductive layer **13A**. For example, the buffer layer **12** may include Tetraethyl Orthosilicate (TEOS).

[0228] A barrier material layer **14A** may be formed over the conductive layer **13A**. The barrier material layer **14A** may include a metal-based material. The barrier material layer **14A** may include a metal, a metal nitride, a metal silicide, or a combination thereof. The barrier material layer **14A** may include titanium nitride, molybdenum, or ruthenium. The barrier material layer **14A** may be formed by an Atomic Layer Deposition (ALD) process, a Physical Vapor Deposition (PVD) process, or a Chemical Vapor Deposition (CVD) process. The barrier material layer **14A** may be formed to have a thickness of approximately 10 to 50 Å. For example, the barrier material layer **14A** may be formed by depositing titanium nitride to a thickness of approximately 20 Å through a physical vapor deposition (PVD) process.

[0229] An oxide semiconductor stack may be formed over the barrier material layer **14A**. For example, the oxide semiconductor stack may include a first interface material layer **15A**, a channel material layer **16A**, and a second interface material layer **17A**.

[0230] The first interface material layer **15A** may be formed over the barrier material layer **14A**.

The first interface material layer **15A** may include a conductive material. The first interface material layer **15A** may include an oxide semiconductor material. The first interface material layer **15A** may contain indium. The first interface material layer **15A** may include an indium-rich oxide semiconductor material. For example, the first interface material layer **15A** may include indium-rich IGZO. The first interface material layer **15A** may be formed to have a thickness of approximately 10 to 50 Å.

[0231] The channel material layer **16A** may be formed over the first interface material layer **15A**. The channel material layer **16A** may include a conductive material. The channel material layer **16A** may include an oxide semiconductor material. The channel material layer **16A** may contain indium. The channel material layer **16A** may include indium gallium zinc oxide (IGZO). The channel material layer **16A** may be formed to have a thickness of approximately 200 to 1000 Å.

[0232] The second interface material layer **17A** may be formed over the channel material layer **16A**. The second interface material layer **17A** may include a conductive material. The second interface material layer **17A** may include an oxide semiconductor material. The second interface material layer **17A** may contain indium. The second interface material layer **17A** may include an indium-rich oxide semiconductor material. For example, the second interface material layer **17A** may include indium-rich IGZO. The second interface material layer **17A** may be formed to have a thickness of approximately 10 to 50 Å.

[0233] As described above, the first interface material layer **15A**, the channel material layer **16A**, and the second interface material layer **17A** may be vertically stacked over the barrier material layer **14A**. The first interface material layer **15A**, the channel material layer **16A**, and the second interface material layer **17A** may all include an oxide semiconductor material. The first interface material layer **15A**, the channel material layer **16A**, and the second interface material layer **17A** may all include indium gallium zinc oxide (IGZO). The first interface material layer **15A** and the second interface material layer **17A** may have a higher indium concentration than that of the channel material layer **16A**. The channel material layer **16A** may be IGZO, and the first interface material layer **15A** and the second interface material layer **17A** may be indium-rich IGZO. Since the first interface material layer **15A** and the second interface material layer **17A** contain a high concentration of indium, the resistance may be reduced compared to that of the channel material layer **16A**, and the seamless interconnection of the channel material layer **16A** may be possible.

[0234] Subsequently, a sacrificial layer **18A** may be formed over the second interface material layer **17A**. The sacrificial layer **18A** may include a stack of different materials. The sacrificial layer **18A** may include silicon nitride.

[0235] Referring to FIG. **16B**, sacrificial lines **18** may be formed over the second interface material layer **17A**. The sacrificial lines **18** may be formed by etching the sacrificial layer **18A**. The sacrificial lines **18** may function to protect the first interface material layer **15A**, the channel material layer **16A**, and the second interface material layer **17A** from a subsequent process. For example, the sacrificial lines **18** may be used as an etching barrier during the etching process of the first interface material layer **15A**, the channel material layer **16A**, and the second interface material layer **17A**.

[0236] Each of the sacrificial lines **18** may include a stack of different materials. The sacrificial lines **18** may include silicon nitride. The etching process of the sacrificial layer **18A** for forming the sacrificial lines **18** may include a double patterning process.

[0237] Subsequently, the second interface material layer **17A**, the channel material layer **16A**, and the first interface material layer **15A** may be etched by using the sacrificial lines **18** as an etching barrier. Further, the barrier material layer **14A** and the conductive layer **13A** may be etched by using the sacrificial lines **18** as an etching barrier.

[0238] A plurality of line structures **19L** and first trenches **19T** may be formed over the buffer layer **12** by a series of the etching processes as described above. Each of the line structures **19L** may include a stack of a bit line **13**, a bit line barrier layer **14**, a first oxide semiconductor interface layer

15B, a channel material layer **16B**, a second oxide semiconductor interface layer **17B**, and a sacrificial line **18**. The bit line barrier layer **14** may be formed by etching the barrier material layer **14A**, and the bit line **13** may be formed by etching the conductive layer **13A**. The first oxide semiconductor interface layer **15B**, the channel material layer **16B**, and the second oxide semiconductor interface layer **17B** may be formed by etching the first interface material layer **15A**, the channel material layer **16A**, and the second interface material layer **17A**, respectively. The first trenches **19T** may be disposed between the line structures **19L**. The stack of the first oxide semiconductor interface layer **15B**, the channel material layer **16B**, and the second oxide semiconductor interface layer **17B** may be referred to as an ‘oxide semiconductor line’.

[0239] Referring to FIG. **16C**, dielectric lines **20** may be formed between the line structures **19L**. The dielectric lines **20** may include a dielectric material. For example, the dielectric lines **20** may include silicon oxide, silicon nitride, silicon carbon oxide (SiCO), a spin-on-dielectric layer, or a combination thereof. The dielectric lines **20** may be formed by depositing a dielectric material to fill the first trenches **19T** between the line structures **19L** and then performing a planarization process of the dielectric material. The dielectric lines **20** may fill the first trenches **19T** respectively. The dielectric lines **20** may be formed by sequentially forming silicon nitride, silicon carbon oxide, and a spin-on-dielectric layer and then performing a planarization process.

[0240] Referring to FIG. **16D**, a portion of the line structures **19L** may be selectively etched to form sacrificial pillars **18P** and oxide semiconductor pillars **21P**. The bit line barrier layer **14** and the bit line **13** may be disposed in the lower portions of the oxide semiconductor pillars **21P**.

[0241] Each of the oxide semiconductor pillars **21P** may include a stack of a first oxide semiconductor interface layer **15**, a channel layer **16**, and a second oxide semiconductor interface layer **17**. The first oxide semiconductor interface layer **15**, the channel layer **16**, and the second oxide semiconductor interface layer **17** may be formed by etching the first oxide semiconductor interface layer **15B**, the channel material layer **16B**, and the second oxide semiconductor interface layer **17B**, respectively. The sacrificial pillars **18P** may be formed over the second oxide semiconductor interface layer **17**.

[0242] Second trenches **22** and **22A** may be formed between the oxide semiconductor pillars **21P**. The bottom surfaces of the second trenches **22** and **22A** may expose the upper surface of the bit line barrier layer **14**. The first trenches **19T** illustrated in FIG. **16B** and the second trenches **22** and **22A** illustrated in FIG. **16C** may intersect with each other. The first trenches **19T** may be deeper than the second trenches **22** and **22A**. In the B-B’ direction, the dielectric lines **20** may be cut by the second trenches **22** and **22A**. Hereinafter, the dielectric lines **20** may be simply referred to as ‘dielectric pillars **20**’. In the A-A’ direction, the dielectric pillars **20** may be disposed between the oxide semiconductor pillars **21P**.

[0243] Each of the oxide semiconductor pillars **21P** may include first to fourth sidewalls SW1 to SW4. The first sidewall SW1 and the second sidewall SW2 of each oxide semiconductor pillar **21P** may be exposed by the second trenches **22**. The third sidewall SW3 and the fourth sidewall SW4 of each oxide semiconductor pillar **21P** may not be exposed by the dielectric pillars **20**. The sacrificial pillar **18P** may also include the exposed sidewalls and the non-exposed sidewalls, just like the oxide semiconductor pillar **21P**.

[0244] The second trenches **22** and **22A** may include the first sub-trench **22** and the second sub-trench **22A**. The first sub-trench **22** and the second sub-trench **22A** may define the space between the neighboring oxide semiconductor pillars **21P**. The width of the first sub-trench **22** may be larger than the width of each second sub-trench **22A**. The first sub-trench **22** may correspond to the first and second spaces S1 and S2 shown in FIGS. **15A** and **15C** and the shield trench ST shown in FIG. **15D**. The second sub-trenches **22A** may correspond to the third space S3 shown in FIG. **15A** and may be referred to as ‘word line trenches’.

[0245] As described above, the space between the neighboring oxide semiconductor pillars **21P** may be asymmetric due to the difference between the width of the first sub-trench **22** and the width

of the second sub-trench **22A**.

[0246] Referring to FIG. **16E**, a gate dielectric layer **23** may be formed over the exposed first and second sidewalls **SW1** and **SW2** of the oxide semiconductor pillars **21P**. The gate dielectric layer **23** may include silicon oxide, silicon nitride, silicon oxynitride, a high-k material, or a combination thereof. The high-k material may include a material having a greater dielectric constant than the dielectric constant of silicon oxide. For example, the high-k material may include a material having a greater dielectric constant than approximately 3.9. For another example, the high-k material may include a material having a dielectric constant greater than approximately 10. For yet another example, the high-k material may include a material having a dielectric constant of approximately 10 to 30. The high-k material may include at least one metallic element. The high-k material may include a hafnium-containing material. The hafnium-containing material may include hafnium oxide, hafnium silicon oxide, hafnium silicon oxynitride, or a combination thereof. According to another embodiment of the present disclosure, the high-k material may include lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, zirconium silicon oxynitride, aluminum oxide, or a combination thereof. Other known high-k materials may be optionally used as the high-k material. The gate dielectric layer **23** may include a metal oxide. The gate dielectric layer **23** may be formed conformally on the first and second sidewalls **SW1** and **SW2** of the oxide semiconductor pillar **21P**. The gate dielectric layer **23** may also conformally cover the exposed sidewalls and upper surface of the sacrificial pillar **18P**. The gate dielectric layer **23** and the first oxide semiconductor interface layer **15** may have the same thickness. According to another embodiment of the present disclosure, the gate dielectric layer **23** may be thinner than the first oxide semiconductor interface layer **15**.

[0247] A word line conductive layer **24A** may be formed over the gate dielectric layer **23**. The word line conductive layer **24A** may be formed conformally over the gate dielectric layer **23**. The word line conductive layer **24A** may include a metal-based material. The word line conductive layer **24A** may include a metal, a metal nitride, or a combination thereof. The word line conductive layer **24A** may include tantalum nitride (TaN), titanium nitride (TiN), tungsten (W), tungsten nitride (WN), or a combination thereof.

[0248] The word line conductive layer **24A** may fully gap-fill the second sub-trenches **22A** over the gate dielectric layer **23**. The word line conductive layer **24A** may be conformally formed over the gate dielectric layer **23**. The word line conductive layer **24A** may partially fill the first sub-trench **22**.

[0249] Spacers **25** may be formed over the word line conductive layer **24A**. The spacers **25** may include an oxide. The spacers **25** may be formed by depositing silicon oxide over the word line conductive layer **24A** and then performing an etch-back process. The upper surfaces of the spacers **25** may be disposed at a lower level than the upper surfaces of the sacrificial pillars **18P**.

[0250] Referring to FIG. **16F**, tapered vertical word lines **24S** may be formed. To form the tapered vertical word lines **24S**, the word line conductive layer **24A** may be selectively etched by using the spacer **25** as an etching barrier. The spacers **25** may function to protect the tapered vertical word lines **24S** during the etching process of the word line conductive layer **24A**. The etching process of the word line conductive layer **24A** may include an etch-back process and a wet etching process that are performed sequentially. Non-tapered vertical word lines may be formed by an etch-back process, and the tapered vertical word lines **24S** may be formed by the subsequent wet etching process. The bottom portion of the tapered vertical word lines **24S** may become thin by the wet etching process. The tapered vertical word lines **24S** may correspond to the main gate **124S** of FIGS. **15A**, **15B**, **15C**, and **15D**, and the bottom portions of the tapered vertical word lines **24S** may correspond to the lower level portion **124L**.

[0251] The tapered vertical word lines **24S** may have a single structure. The tapered vertical word lines **24S** may be disposed on one side of the oxide semiconductor pillars **21P**.

[0252] While the tapered vertical word lines **24S** are formed, a common gate, i.e., a shield gate

24G, may be formed. The shield gate **24G** may be formed in the second sub-trench **22A**.

[0253] The shield gate **24G** may correspond to the shield gate **124G** of FIGS. **15A**, **15B**, **15C**, and **15D**.

[0254] Referring to FIG. **16G**, the spacers **25** of FIG. **16F** may be removed. Each of the tapered vertical word lines **24S** may be a single structure and may be formed on one sidewall among the first and second sidewalls of the oxide semiconductor pillars **21P**. Referring to FIGS. **15A**, **15B**, **15C**, and **15D**, the tapered vertical word lines **24S** may extend in the second direction **D2**, and the bit lines **13** may extend in the first direction **D1**. The oxide semiconductor pillars **21P** and the dielectric pillars **20** may be alternately disposed in the second direction **D2**. The tapered vertical word lines **24S** may extend along the profile of the exposed sidewalls of the oxide semiconductor pillars **21P** and the dielectric pillars **20**.

[0255] As described above, the space between the neighboring oxide semiconductor pillars **21P** may be asymmetric due to the difference between the width of the first sub-trench **22** and the width of each second sub-trench **22A**. Accordingly, when the word line conductive layer **24A** is deposited, the second sub-trenches **22A** may be completely buried and the first sub-trench **22** may be deposited in the form of a spacer. As a result, the shield gate **24G** may have a single buried structure, and only the main gate **124S** may be separated.

[0256] Since the shield gate **24G** is shared, the space of the main gate **24S** may be widened, thereby securing additional process margin.

[0257] Referring to FIG. **16H**, an inter-layer dielectric layer **26** may be formed over the tapered vertical word line **24S**. The inter-layer dielectric layer **26** may be planarized to expose the upper surfaces of the sacrificial pillars **18P** of FIG. **16G**. The inter-layer dielectric layer **26** may include silicon oxide, such as a spin-on-dielectric (SOD) layer. The sacrificial pillars **18P** may serve as a stop layer during the planarization process of the inter-layer dielectric layer **26**.

[0258] Subsequently, the sacrificial pillars **18P** may be selectively removed. As a result, hole-shaped recesses **18R** may be formed. The hole-shaped recesses **18R** may selectively expose the surfaces of the second oxide semiconductor interface layers **17**. The sacrificial pillars **18P** may be removed by performing a wet etching process.

[0259] Referring to FIG. **16I**, first contact plugs **27** filling the hole-shaped recesses **18R** may be formed. The first contact plugs **27** may directly contact the second oxide semiconductor interface layers **17**. The first contact plugs **27** may include a metal, a metal nitride, or a combination thereof. For example, the first contact plugs **27** may be formed by depositing titanium nitride to fill the hole-shaped recesses **18R** of FIG. **16H**, and then planarizing the titanium nitride to expose the surface of the inter-layer dielectric layer **26**.

[0260] Subsequently, referring to FIG. **2J**, second contact plugs **28** may be formed over the first contact plugs **27**.

[0261] Subsequently, referring to FIG. **2K**, a spacer material layer **29** may be formed between the second contact plugs **28**. The spacer material layer **29** may include silicon nitride. The spacer material layer **29** may be formed between the neighboring second contact plugs **28**. A capacitor **30** may be formed over each of the second contact plugs **28**.

[0262] This technology may be able to reduce the space between the neighboring cells by forming a main gate and a shield gate each having a single structure, thereby reducing the cell area and increasing the degree of integration or securing the process margin.

[0263] According to the embodiment of the present disclosure, since a bit line and a transistor are sequentially formed by using a depositable channel material, procedural difficulty may be reduced.

[0264] According to the embodiment of the present disclosure, the degree of integration may be increased by stacking a memory cell array without wafer bonding, and the degree of integration may be improved through a peripheral-under-cell (PUC) structure where a memory cell array is formed over a peripheral circuit portion.

[0265] According to the embodiment of the present disclosure, a metal oxide, particularly IGZO,

may be used as a channel material to suppress gate-induced drain leakage and junction leakage to improve retention characteristics.

[0266] The effects desired to be obtained in the embodiments of the present disclosure are not limited to the effects mentioned above, and other effects not mentioned above may also be clearly understood by those of ordinary skill in the art to which the present disclosure pertains from the description above.

[0267] While the embodiments of the present disclosure have been described with respect to specific embodiments, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the present disclosure as defined in the following claims. Furthermore, the embodiments may be combined to form additional embodiments.

Claims

1. A semiconductor device comprising: a first pillar and a second pillar spaced apart from each other by a shield trench, each pillar including an inner side that defines inner sidewalls of the shield trench and an outer side that faces the inner side; a shield gate formed in the shield trench; a first main gate formed on the outer side of the first pillar; a second main gate formed on the outer side of the second pillar; a bit line formed on lower portions of the first and second pillars; and a capacitor formed on an upper portion of each of the first and second pillars.
2. The semiconductor device of claim 1, wherein each of the first and second main gates has a tapered shape.
3. The semiconductor device of claim 2, wherein each of the first and second main gates includes: a first sidewall of a straight profile; and a second sidewall of a tapered profile.
4. The semiconductor device of claim 3, wherein the first sidewalls of the first and second main gates are formed on the outer sides of the first and second pillars, respectively.
5. The semiconductor device of claim 1, wherein each of the first and second main gates includes: an upper level portion; and a lower level portion which is thinner than the upper level portion.
6. The semiconductor device of claim 1, further comprising: a gate dielectric layer formed between the outer sides of the first and second pillars and the first and second main gates; and a shield gate dielectric layer formed between the shield gate and the inner sides of the first and second pillars.
7. The semiconductor device of claim 1, wherein each of the first and second pillars includes: a first oxide semiconductor interface layer formed over the bit line; a second oxide semiconductor interface layer formed below the capacitor; and an oxide semiconductor channel layer between the first oxide semiconductor interface layer and the second oxide semiconductor interface layer.
8. The semiconductor device of claim 7, wherein each of the oxide semiconductor channel layer, the first oxide semiconductor interface layer, and the second oxide semiconductor interface layer includes an oxide semiconductor material, and each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes an oxide semiconductor material having a lower resistance than the oxide semiconductor channel layer.
9. The semiconductor device of claim 7, wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes a metal-rich oxide semiconductor material having a higher metal content than the oxide semiconductor channel layer.
10. The semiconductor device of claim 7, wherein the oxide semiconductor channel layer includes indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO) or zinc tin oxide (ZTO), wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes indium-rich IGZO, and wherein the indium-rich IGZO includes IGZO whose indium content is higher than gallium and zinc contents.
11. A semiconductor device comprising: a trench; a first row array of first oxide semiconductor pillars and a second row array of second oxide semiconductor pillars spaced apart from each other

by the trench, each pillar including an inner side that defines inner sidewalls of the trench and an outer side that faces the inner side; a common gate extending along the first row array and the second row array, and formed in the trench while being formed on the inner sides of the first and second oxide semiconductor pillars; a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and a second gate extending along the outer sides of the second oxide semiconductor pillars of the second array.

12. The semiconductor device of claim 11, wherein the first and second oxide semiconductor pillars of the first and second row arrays form a column array in which the first oxide semiconductor pillars and the second oxide semiconductor pillars are alternately disposed in a direction intersecting with the first and second row arrays.

13. The semiconductor device of claim 12, further comprising: a bit line formed on lower portions of the first and second oxide semiconductor pillars of the column array; and capacitors formed in upper portions of the first and second oxide semiconductor pillars of the row arrays and the column array.

14. The semiconductor device of claim 11, wherein each of the first and second gates has a tapered shape.

15. The semiconductor device of claim 11, wherein each of the first and second gates includes: a first sidewall of a straight profile; and a second sidewall of a tapered profile.

16. The semiconductor device of claim 15, wherein the first sidewalls of the first and second gates are formed on the outer sides of the first and second oxide semiconductor pillars, respectively.

17. The semiconductor device of claim 11, further comprising: a gate dielectric layer formed between the outer sides of the first and second oxide semiconductor pillars and the first and second gates; and a common gate dielectric layer formed between the common gate and the inner sides of the first and second oxide semiconductor pillars.

18. The semiconductor device of claim 11, wherein each of the first and second oxide semiconductor pillars includes: a first oxide semiconductor interface layer; a second oxide semiconductor interface layer; and an oxide semiconductor channel layer between the first oxide semiconductor interface layer and the second oxide semiconductor interface layer.

19. The semiconductor device of claim 18, wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes a metal-rich oxide semiconductor material having a higher metal content than the oxide semiconductor channel layer.

20. The semiconductor device of claim 18, wherein the oxide semiconductor channel layer includes indium gallium zinc oxide (IGZO), indium tin zinc oxide (ITZO), or zinc tin oxide (ZTO), wherein each of the first oxide semiconductor interface layer and the second oxide semiconductor interface layer includes indium-rich IGZO, and wherein the indium-rich IGZO includes IGZO whose indium content is higher than gallium and zinc contents.

21. A method for fabricating a semiconductor device, the method comprising: forming a bit line; forming a first row array of first oxide semiconductor pillars and a second row array of second oxide semiconductor pillars spaced apart from each other by a trench, each pillar including an inner side that defines inner sidewalls of the trench and an outer side that faces the inner side over the bit line; forming a common gate extending along the first row array and the second row array on the inner sides of the first and second oxide semiconductor pillars; forming a first gate extending along the outer sides of the first oxide semiconductor pillars of the first row array; and forming a second gate extending along the outer sides of the second oxide semiconductor pillars of the second row array.

22. The method of claim 21, wherein forming the first row array of the first oxide semiconductor pillars and the second row array of the second oxide semiconductor pillars includes: forming an oxide semiconductor stack; forming a plurality of oxide semiconductor lines by etching the oxide semiconductor stack; forming dielectric lines between the oxide semiconductor lines; and etching the oxide semiconductor lines to form the first and second oxide semiconductor pillars.

23. The method of claim 22, wherein the oxide semiconductor stack includes a first oxide semiconductor interface layer, an oxide semiconductor channel layer, and a second oxide semiconductor interface layer that are sequentially stacked.

24. The method of claim 21, wherein forming the common gate, the first gate and the second gate includes: forming a preliminary conductive layer between the oxide semiconductor pillars; dry-etching the preliminary conductive layer to form a non-tapered vertical conductive layer; and wet-etching a lower region of the non-tapered vertical conductive layer to form a vertical conductive layer having a tapered shape.

25. The method of claim 21, wherein each of the first and second oxide semiconductor pillars includes an oxide semiconductor material.

26. The method of claim 21, further comprising: forming contact plugs over the first and second oxide semiconductor pillars of the row array; and forming capacitors over the contact plugs.
